Astronomy & Astrophysics manuscript no. paper
July 15, 2026

Are galaxies in compact groups special?

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ABSTRACT

Context. Compact groups of galaxies combine high projected densities with low velocity dispersions, but many are embedded in, or projected onto, larger structures. Their evolutionary significance depends on whether their member galaxies remain distinct from those of comparable regular groups, not only from the field.

Aims. We test whether galaxies in compact groups of four members (CG_4) remain special against regular groups of four members (RG_4), the four brightest galaxies of richer regular groups ($\mathrm{Control}_{4\mathrm{B}}$), and projected compact cores formed by the brightest group galaxy (BGG) and its three nearest projected companions ($\mathrm{Control}_{4\mathrm{C}}$). We ask which observable carries any residual signal.

Methods. From our previously constructed samples, we retain 62 CG_4 systems after removing split systems. We combine Galaxy Zoo vote-fraction morphologies, Sloan Digital Sky Survey stellar masses and specific star-formation rates, optical colours, emission-line diagnostics, and seeing-corrected half-light radii (with Petrosian radii as a cross-check), analysed with covariate-adjusted, group-clustered binomial models and one-to-one matched controls.

Results. CG_4 galaxies, especially satellites, are more often smooth/elliptical and less often spiral than their regular-group analogues; this signal survives adjustment for stellar mass, redshift, rank, and group-scale quantities, and is recovered in matched comparisons. The star-formation deficit of CG_4 satellites is weaker and not comparably robust after adjustment and matching, and star-forming CG_4 galaxies show no robust offset from the main sequence. Colours, fractions of active galactic nuclei (AGN), magnitude gaps, and Hα diagnostics do not support instantaneous-quenching, fossil-group, or AGN-driven explanations. Conditioning on a projected pairwise tidal index attenuates the morphology signal, indicating that the contrast is concentrated in the most extreme local projected configurations. No robust CG_4 size offset at fixed stellar mass is established; the matched-pair preference for smaller sizes is control-dependent.

Conclusions. Compact groups host a more morphologically evolved population, consistent with accumulated transformation in dense group cores and/or preprocessing, rather than a uniquely isolated compact-group quenching mode at the present epoch.

Key words. galaxies: groups: general – galaxies: evolution – galaxies: interactions – galaxies: star formation – galaxies: elliptical and lenticular, cD – methods: statistical

1. Introduction

Compact groups of galaxies are useful laboratories for environmental evolution because they combine high projected galaxy densities with relatively low velocity dispersions (Hickson 1982; Hickson et al. 1992). Mendes de Oliveira & Hickson (1994) provided the classic report of an excess of early-type galaxies in compact groups relative to the field. In such configurations, repeated tidal encounters, dynamical friction, mergers, gas stripping or consumption, and disk heating can plausibly accelerate the transformation of galaxies compared with more diffuse environments (Barnes 1985, 1989; Verdes-Montenegro et al. 2001). This expectation is closely connected to the broader morphology-density and star-formation-environment relation, in which galaxy structure and star-formation activity vary systematically with local density (Dressler 1980). Multi-wavelength studies of compact-group galaxies likewise report altered infrared, UV, star-formation, and population properties relative to looser environments (Johnson et al. 2007; Walker et al. 2010; Tzanavaris et al. 2010; Bitsakis et al. 2010, 2011; Coenda et al. 2012; Plauchu-Frayn et al. 2012).

The interpretation of compact groups is nevertheless ambiguous. Projected selection can include chance alignments, and many Hickson-like systems need not be isolated bound entities. Simulations and redshift-space analyses have long shown that apparent compact groups may be transient alignments, dense substructures, or cores embedded in looser groups and richer haloes (Mamon 1986; Díaz-Giménez & Mamon 2010; Díaz-Giménez et al. 2012; Díaz-Giménez et al. 2020; Zheng & Shen 2021; Taverna et al. 2023). The relevant observational question is therefore not only whether compact-group galaxies differ from field galaxies, but whether they remain different

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from galaxies occupying regular-group analogues selected with comparable luminosity, redshift, and membership constraints.

Recent structural work strengthens this motivation. Using S-PLUS imaging and single-Sérsic measurements, Montaguth et al. (2023, 2025, 2026a,b) identified transition galaxies and trends in the R_e–Sérsic-index plane that point to structural transformation in compact groups, with stronger signatures in non-isolated compact groups and host structures. These studies suggest that compact groups can mark sites of cumulative structural processing. They do not, however, by themselves answer whether compact-group galaxies remain distinct after comparison with regular-group cores selected from the same parent catalogues.

Our previous work (Tricottet et al. 2025) addressed the question of whether compact groups are special as dynamical systems. We found that, once the direct consequences of the compact-group selection are accounted for, compact groups of four galaxies have group-scale properties broadly similar to those of regular groups of four galaxies and to the cores of richer regular groups. This motivates the complementary question addressed here: even if compact groups are not strongly distinct as systems, are their member galaxies individually different? We therefore use the same compact-group and regular-group framework to compare galaxies rather than groups, controlling for stellar mass, redshift, galaxy rank, and available group-scale properties. We then identify which observable, if any, carries the residual compact-group signal: star-formation class, Galaxy Zoo morphology, optical colour, AGN-like activity, fossil-like magnitude gaps, projected phase-space position, or local projected tidal density.

The paper is organized as follows. Section 2 describes the samples, the sSFR and morphology classifications, and the statistical hierarchy. Sections 3 and 4 present baseline population differences and group-scale diagnostics. Section 5 tests which signals survive adjusted and matched comparisons and examines secondary diagnostics. Section 6 discusses the physical interpretation of the morphology signal, and Section 7 summarizes the conclusions.

2. Data and classifications

2.1. Samples

We use the compact-group and control samples constructed in our previous study (Tricottet et al. 2025). We briefly recall their construction here. Galaxy properties are taken from the SDSS DR12 catalogue of Tempel et al. (2017), while regular groups are drawn from the SDSS DR7 group catalogue of Lim et al. (2017); the two catalogues were matched through SDSS object identifiers to define the Lim–Tempel working catalogue. Compact groups are selected from the modified Hickson compact-group catalogue of Díaz-Giménez et al. (2018). Starting from the reliable HMCGs with exactly four members, we retain only systems in which all galaxies are present in the Lim–Tempel catalogue and impose the volume- and luminosity-complete limits $0.005 < z_{\rm BGG} < 0.0452$ and $M_{r,\rm BGG} \leq -21.81$, with all 4 galaxies lying within 3 magnitudes of the brightest group galaxy. This defines the parent $\rm CG_4$ sample of 78 compact groups. The regular-group comparison samples are built from Lim–Tempel groups satisfying the same redshift and BGG-luminosity limits, with at least 4 members and at least 3 satellites within 3 magnitudes of the BGG. From this parent control catalogue, we define $\rm RG_4$ groups as regular groups with exactly four members, excluding systems identical to a $\rm CG_4$, and we additionally use the $\rm Control_{4B}$ and $\rm Control_{4C}$ samples, formed respectively from the 4 brightest galaxies and from the BGG plus its 3 closest projected companions in each parent regular group. The current input catalogues are treated as fixed selected-core catalogues in this paper; after the explicit Lim group 3688 removal, they contain 698 $\rm Control_{4B}$ cores and 751 $\rm Control_{4C}$ cores. These counts differ because the luminosity-ranked and projected-distance-ranked core definitions select different galaxies and therefore retain different Lim groups. In contrast to our previous study, we remove CGs classified as *split* following Zheng & Shen (2021). This reduces the compact-group sample from 78 to 62 groups. We also remove the systems associated with Lim group 3688 from our $\rm Control_{4B}$ and $\rm Control_{4C}$ samples. This group contains 6 galaxies with redshifts between 0.032 and 0.034, and 1 with a redshift of 0.048. This group appears to be spurious, and the outlying galaxy, although not the BGG, is selected in both our original $\rm Control_{4B}$ and $\rm Control_{4C}$ samples, significantly affecting the group velocity dispersion (1979 and 1981 $\rm km\,s^{-1}$, respectively).

The three regular-group controls serve different physical purposes. $\rm RG_4$ contains true regular groups with exactly four selected members and is the most direct four-member regular-group comparison. $\rm Control_{4B}$ contains the four brightest galaxies in richer regular groups and tests whether the $\rm CG_4$ population resembles the luminous core of an ordinary group. $\rm Control_{4C}$ contains the BGG plus the three closest projected companions and is the most compact-core-like ordinary-group analogue. $\rm CG_4$ is therefore tested against both equal-membership groups and regular-group cores, rather than against a single control definition.

We use SDSS DR16 (Ahumada et al. 2020) to retrieve stellar masses, star formation rates (SFRs; Brinchmann et al. 2004; Kauffmann et al. 2003b), and Galaxy Zoo morphologies (Lintott et al. 2008, 2011). We specifically extract the columns `sfr_tot_p50`, `specsfr_tot_p50`, and `lgm_tot_p50` from the `galSpecExtra` table, and `p_el_debiased` and `p_cs_debiased` from the `zooSpec` table. In the SDSS query, `galSpecExtra`, `galSpecLine`, and `zooSpec` are joined to `SpecObj` by `specObjID`, while the photometric object is linked through `SpecObj.bestObjID=PhotoObj.objID`; the queried quantities are then merged onto the CG and control catalogue rows by their stored `objid`.

Throughout, 'SDSS selection' denotes the reference sample returned by the query of Appendix A: all spectroscopic galaxies with $0.005 \leq z \leq 0.0452$, extinction-corrected Petrosian $r \leq 17.77$, a valid MPA-JHU stellar mass, and Galaxy Zoo classifications. It serves as a field-dominated reference population, not as a control sample.

All projected distances and derived quantities adopt the Planck 2015 cosmology (Planck Collaboration et al. 2016), consistently with our previous paper (Tricottet et al. 2025).

2.2. sSFR classification

We classify galaxies from SDSS total sSFRs using the query reported in Appendix A (Listing 1). In the MPA-JHU tables, `specsfr_tot_p50` = −9999 flags galaxies with-

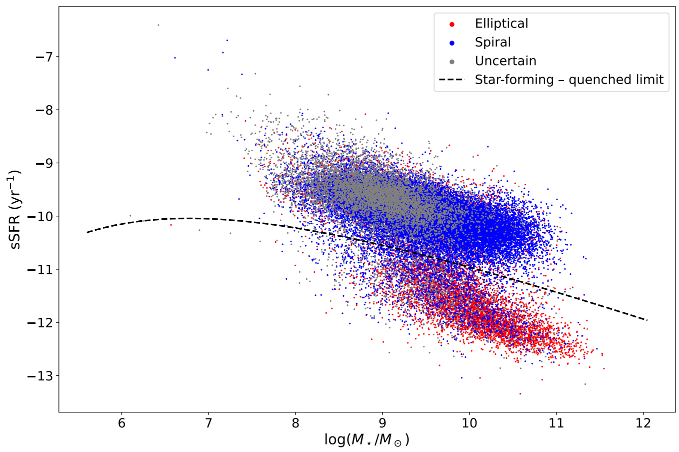

Fig. 1. sSFR versus stellar mass, the star-forming/quenched limit, and Galaxy Zoo morphologies for the SDSS selection, excluding galaxies without sSFR estimates.

out a valid sSFR estimate rather than measured zero star formation; we therefore treat such galaxies as *missing data*: they are excluded from the star-formation classification, from every sSFR-based figure and fraction, and are reported as counts only (Table 1). All quenched and star-forming fractions quoted in this paper are computed among classified galaxies only. The missing-sSFR incidence is markedly higher in CG$_4$ (7% of members, and 10% of BGGs) than in the field selection (1%), plausibly reflecting the crowded-field spectroscopy and deblending issues discussed in Sect. 6.4 rather than a physical population. Galaxies with a measured sSFR are split into quenched and star-forming classes with a two-component GMM in the $\log M_\star$-$\log \mathrm{sSFR}$ plane. BPT/AGN handling and the exact GMM boundary definition are given in Appendix B.

2.3. Morphology classification

Morphologies are determined via the Galaxy Zoo citizen-science decision tree, in which each galaxy image receives multiple independent volunteer classifications that are aggregated into debiased vote fractions (Lintott et al. 2008, 2011). Those fractions are provided in the `zooSpec` table of the SDSS database through the `p_el_debiased` and `p_cs_debiased` columns. We classify a galaxy as "elliptical/smooth" if `p_el_debiased` is greater than 0.5, and as "spiral/features" if `p_cs_debiased` is greater than 0.5. All other cases are classified as "uncertain". These are vote-fraction classes, not a full structural decomposition. In particular, the catalogue used here does not provide a clean E/S0 separation, so references below to elliptical or early-type galaxies should be read as Galaxy Zoo smooth/early-type proxies rather than as a bulge-disc structural classification.

2.4. Galaxy sizes

Our primary galaxy-size measure is the circular half-light radius of the pure Sérsic fit in the r band, $R_{\mathrm{chl,r}}$, from the bulge+disk decomposition catalogue of Simard et al. (2011). These are PSF-convolved GIM2D fits with re-derived sky levels and deblending designed with crowded fields in mind, which is the relevant regime for compact-group photometry, so the fitted sizes are corrected for seeing. The catalogue is keyed to SDSS DR7 identifiers; we therefore bridge our DR16 objects to their DR7 counterparts through the SkyServer `SpecDR7` cross-match table. Two guards protect this bridge: rows in which the Simard redshift differs from the catalogue redshift by more than 0.005 are treated as wrong matches and their structural values are dropped (15 rows), and rows in which two distinct catalogue galaxies of the same group resolve to a single DR7 identifier are treated as blended DR7 detections and dropped (0 rows). The Simard sizes are tabulated in kpc under the catalogue's own cosmology; we convert them back to angular sizes with the catalogue plate scale and then to proper kpc with the same Planck 2015 conversion as every other projected quantity in this paper (Sect. 2.1).

As a secondary size measure we use the SDSS Petrosian radii `petroR50_r` and `petroR90_r`, which also define the concentration index $C = R_{90,r}/R_{50,r}$. Petrosian radii are not corrected for seeing, and the SDSS Petrosian aperture misses a profile-dependent fraction of the flux, roughly 20% for an $n = 4$ profile, which biases $R_{50,r}$ low for concentrated galaxies (Blanton et al. 2001; Graham et al. 2005) – precisely the smooth/early-type-like population in which CG$_4$ is enhanced. The Petrosian measure is nevertheless retained because it is model-independent, essentially fully available, and provides the like-for-like bridge to the compact-group size study of Coenda et al. (2012). All sizes are required to lie between 0.1 and 50 kpc after re-conversion; Sérsic fits pegged at the GIM2D bounds ($n_g \leq 0.55$ or $n_g \geq 7.9$; 63 galaxies) are excluded from the primary sample, and Petrosian entries must satisfy $R_{50,r} > 0$, $R_{90,r} > R_{50,r}$, and a finite positive uncertainty. The completeness of both measures, and its difference between CG$_4$ and the controls, is audited in Sect. 5.9.

2.5. Statistical strategy

The analyses are organized hierarchically. Raw tables and exact tests provide descriptive baseline comparisons. Fisher or Barnard exact tests are used for two-sample categorical comparisons depending on the table structure, while rank-sum or bootstrap comparisons are used for distributional and median differences. Spearman and Pearson correlations are used only as diagnostics of monotonic or linear trends; when the coefficients are small, we interpret them as weak even if the p-values are formally small. Holm or Benjamini–Hochberg corrections are applied where multiple related tests are performed. The primary robustness layer consists of mass-, rank-, and environment-adjusted binomial models with group-clustered standard errors, followed by one-to-one matched control comparisons. Colours, AGN-like fractions, Hα equivalent widths, magnitude gaps, crossing-time trends, projected phase space, and the pairwise tidal index are secondary diagnostics. The main conclusions below rely on signals that survive the adjusted and matched analyses, not on isolated exploratory tests.

3. Baseline population differences

3.1. Raw sSFR fractions

The sSFR status for each sample is shown in Table 1. In the raw all-galaxy comparison, Fisher's exact test gives star-forming-fraction p-values of $p = 0.481$ versus the Control$_{4\mathrm{B}}$

Table 1. Number of galaxies in each sSFR class for each sample. Galaxies without a valid sSFR estimate ('No sSFR') are excluded from the classification: their percentages are quoted with respect to all galaxies of the (sub)sample, while the quenched and star-forming percentages are quoted with respect to classified galaxies only.

Sample	**No sSFR**	**Quenched**	**Star-forming**
CG_4	18 (7%)	146 (63%)	84 (37%)
BGG	*6 (10%)*	*45 (80%)*	*11 (20%)*
Satellites	*12 (6%)*	*101 (58%)*	*73 (42%)*
$Control_{4B}$	180 (6%)	1593 (61%)	1019 (39%)
BGG	*58 (8%)*	*533 (83%)*	*107 (17%)*
Satellites	*122 (6%)*	*1060 (54%)*	*912 (46%)*
$Control_{4C}$	179 (6%)	1551 (59%)	1086 (41%)
BGG	*59 (8%)*	*538 (83%)*	*107 (17%)*
Satellites	*120 (6%)*	*1013 (51%)*	*979 (49%)*
RG_4	5 (2%)	101 (46%)	118 (54%)
BGG	*3 (5%)*	*38 (72%)*	*15 (28%)*
Satellites	*2 (1%)*	*63 (38%)*	*103 (62%)*
SDSS selection	*634 (1%)*	*7705 (15%)*	*44192 (85%)*

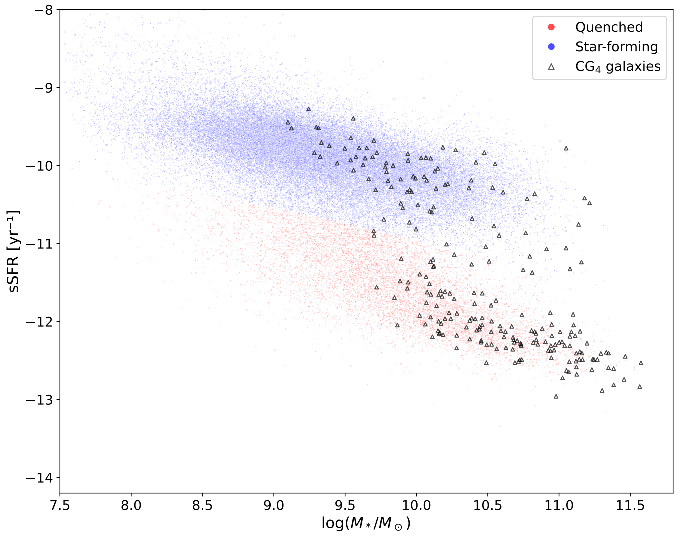

Fig. 2. sSFR and morphologies of galaxies in CG_4 and our SDSS selection. Galaxies without a valid sSFR estimate are excluded from this figure.

Table 2. Number of galaxies in each Galaxy Zoo vote-fraction morphology proxy for each sample.

Sample	**Elliptical/ smooth**	**Spiral/ features**	**Uncertain**
CG_4	124 (50%)	86 (35%)	38 (15%)
$Control_{4B}$	1106 (40%)	1266 (45%)	420 (15%)
$Control_{4C}$	1157 (41%)	1140 (40%)	519 (18%)
RG_4	60 (27%)	129 (58%)	35 (16%)
SDSS selection	*10091 (19%)*	*34715 (66%)*	*7725 (15%)*

sample, $p = 0.184$ versus $Control_{4C}$, and $p = 3 \times 10^{-4}$ versus RG_4. We treat these raw tests as descriptive because they do not adjust for stellar mass, rank, or group-scale differences.

The same comparison restricted to BGGs and satellites is also shown in Table 1. For BGGs, the p-values obtained with two-sided Fisher exact test are $p = 0.578$ against $Control_{4B}$, $p = 0.576$ against $Control_{4C}$ and $p = 0.37$ against RG_4. For satellites, the corresponding p-values are $p = 0.302$ against $Control_{4B}$, $p = 0.069$ against $Control_{4C}$ and $p = 2.3 \times 10^{-4}$ against RG_4. The satellite deficit is clearest in the raw comparison with RG_4 and is revisited below with adjusted and matched analyses.

3.2. Raw morphology fractions

The morphologies obtained for each sample are shown in Table 2. Barnard's two-sided exact tests were used to compare the fraction of each morphology between CG_4 and each control sample. The resulting p-values are $p = 1.4 \times 10^{-3}$ for ellipticals and $p = 1.2 \times 10^{-3}$ for spirals versus the $Control_{4B}$ sample, $p = 6.4 \times 10^{-3}$ and $p = 0.074$ versus $Control_{4C}$ and $p < 10^{-4}$ and $p < 10^{-4}$ versus RG_4. The galaxies in CG_4 systems, which we identified as cores of richer groups in Tricottet et al. (2025), show a robust raw excess of elliptical/smooth classifications and a deficit of spiral/features classifications. Figure 2 displays sSFR versus stellar mass for galaxies in the SDSS selection and in CG_4.

3.3. Morphology versus sSFR

Among star-forming CG_4 galaxies, 58 are classified as spiral (84.1% after removing galaxies with uncertain morphology) and 11 are classified as elliptical (15.9% after removing galaxies with uncertain morphology). The corresponding numbers are 810 spiral galaxies (88.4%) and 106 elliptical galaxies (11.6%) for $Control_{4B}$, 773 spiral galaxies (84.3%) and 144 elliptical galaxies (15.7%) for $Control_{4C}$, and 91 spiral galaxies (91%) and 9 elliptical galaxies (9%) for RG_4. The probabilities that these fractions are drawn from the same distribution, estimated with Barnard two-sided exact test, are $p = 0.283$ against $Control_{4B}$, $p = 0.964$ against $Control_{4C}$, and $p = 0.186$ against RG_4. We therefore find neither a significant excess nor a deficit of star-forming elliptical galaxies in CG_4 relative to any control sample.

When restricting the comparison to BGGs, we find 36 elliptical BGGs (58.1% of the BGGs) and 17 spiral BGGs (27.4%) in CG_4. The corresponding numbers are 405 elliptical BGGs (58.0%) and 195 spiral BGGs (27.9%) in $Control_{4B}$, 409 elliptical BGGs (58.1%) and 195 spiral BGGs (27.7%) in $Control_{4C}$, and 21 elliptical BGGs (37.5%) and 26 spiral BGGs (46.4%) in RG_4. The p-values obtained with two-sided Fisher exact test are 1.00 against $Control_{4B}$, 1.00 against $Control_{4C}$ and 0.03 against RG_4. Thus, BGG morphologies differ between CG_4 and RG_4: CG_4 BGGs have a higher elliptical fraction than RG_4 BGGs.

3.4. Population medians by sSFR class

Tables 3–5 summarize the typical stellar masses, absolute magnitudes, and sSFRs of quenched and star-forming galaxies, including splits by the sSFR class of the BGG. These medians are descriptive and mainly show that star-formation class is entangled with stellar mass and luminosity, motivating the adjusted models below.

Table 3. Median values of galaxy properties for quenched and star-forming populations in each sample. Uncertainties are estimated by bootstrapping (10 000 iterations). sSFR is in yr^{-1}, and stellar mass is in $M_{\odot}$.

Sample	Quantity	Quenched	Star-forming
CG4	$\log_{10}$ sSFR	−12.08 ± 0.03	−10.19 ± 0.07
	M_r	−21.16 ± 0.14	−20.4 ± 0.14
	$\log M_{\star}$	10.64 ± 0.07	10.05 ± 0.05
Control4B	$\log_{10}$ sSFR	−12.1 ± 0.01	−10.31 ± 0.02
	M_r	−21.52 ± 0.04	−20.75 ± 0.04
	$\log M_{\star}$	10.79 ± 0.01	10.24 ± 0.02
Control4C	$\log_{10}$ sSFR	−11.93 ± 0.02	−10.18 ± 0.02
	M_r	−20.81 ± 0.06	−19.95 ± 0.04
	$\log M_{\star}$	10.47 ± 0.03	9.84 ± 0.02
RG4	$\log_{10}$ sSFR	−11.91 ± 0.07	−10.16 ± 0.07
	M_r	−21.11 ± 0.25	−20.32 ± 0.12
	$\log M_{\star}$	10.62 ± 0.08	9.97 ± 0.09

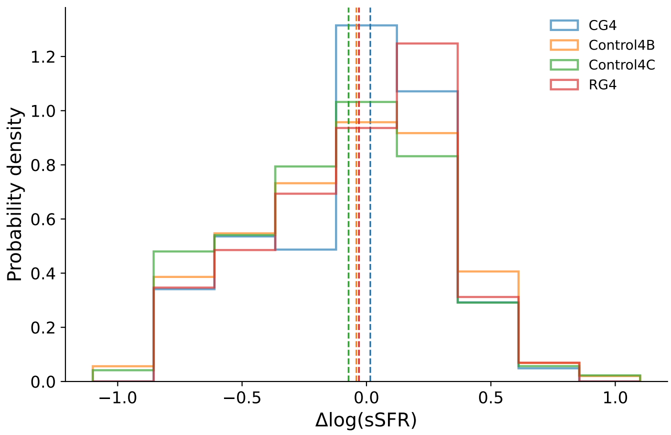

Fig. 3. Main-sequence residual distributions for the star-forming galaxies in each sample.

Table 4. Median values of various quantities for galaxies according to the sSFR class of their BGG (BGG included). Uncertainties are estimated by bootstrapping (10 000 iterations). sSFR is in yr^{-1}, and stellar mass is in $M_{\odot}$.

Sample	Quantity	BGG quenched	BGG SF
CG4	$\log_{10}$ sSFR	−11.53 ± 0.09	−10.85 ± 0.29
	M_r	−20.93 ± 0.11	−20.77 ± 0.32
	$\log M_{\star}$	10.46 ± 0.06	10.32 ± 0.12
Control4B	$\log_{10}$ sSFR	−11.52 ± 0.03	−10.86 ± 0.06
	M_r	−21.25 ± 0.03	−21.12 ± 0.07
	$\log M_{\star}$	10.61 ± 0.02	10.46 ± 0.04
Control4C	$\log_{10}$ sSFR	−11.33 ± 0.04	−10.65 ± 0.07
	M_r	−20.45 ± 0.05	−20.59 ± 0.08
	$\log M_{\star}$	10.24 ± 0.02	10.19 ± 0.06
RG4	$\log_{10}$ sSFR	−11.14 ± 0.19	−10.58 ± 0.16
	M_r	−20.69 ± 0.14	−20.74 ± 0.28
	$\log M_{\star}$	10.32 ± 0.07	10.19 ± 0.12

Table 5. Median values of various quantities for galaxies according to the sSFR class of their BGG (satellites only). Uncertainties are estimated by bootstrapping (10 000 iterations). sSFR is in yr^{-1}, and stellar mass is in $M_{\odot}$.

Sample	Quantity	BGG quenched	BGG SF
CG4	$\log_{10}$ sSFR	−11.23 ± 0.16	−10.92 ± 0.48
	M_r	−20.45 ± 0.13	−20.34 ± 0.18
	$\log M_{\star}$	10.22 ± 0.05	10.09 ± 0.06
Control4B	$\log_{10}$ sSFR	−11.25 ± 0.06	−10.91 ± 0.13
	M_r	−20.9 ± 0.03	−20.75 ± 0.06
	$\log M_{\star}$	10.43 ± 0.02	10.3 ± 0.03
Control4C	$\log_{10}$ sSFR	−10.99 ± 0.04	−10.62 ± 0.09
	M_r	−19.82 ± 0.04	−20.05 ± 0.09
	$\log M_{\star}$	9.92 ± 0.02	9.93 ± 0.06
RG4	$\log_{10}$ sSFR	−10.82 ± 0.19	−10.61 ± 0.22
	M_r	−20.23 ± 0.16	−20.3 ± 0.13
	$\log M_{\star}$	10.09 ± 0.08	10 ± 0.13

3.5. Main sequence

We model the star-forming main sequence as a function of stellar mass using polynomial orders 1–4; the polynomial-order comparison is shown in Appendix Fig. C.4. The diagnostic shown there is the residual RMS scatter, not a formal reduced χ^2, because no per-galaxy uncertainty model is used. We adopt order 2 because it is the lowest-order model with the smallest five-fold cross-validated RMS scatter (0.30523 dex, compared with 0.30560 dex for order 1). We then use order 2 to compute the residual of each star-forming galaxy with respect to the main sequence, and show their distributions in Fig. 3. A positive residual means higher sSFR at fixed stellar mass relative to the fitted star-forming main sequence.

In the comparisons below, Δ is the CG_4-minus-control median residual; positive values mean that CG_4 star-forming galaxies lie above the fitted main sequence relative to the control. Each quoted interval is the central 68% (16th–84th percentile) bootstrap interval on the median difference, and each p is the two-sided bootstrap sign-crossing probability for the same difference (10 000 resamples). The 68% interval and the two-sided p answer slightly different questions and can therefore appear to disagree near the significance boundary: a 68% interval that excludes zero is compatible with p slightly above 0.05, as for $Control_{4B}$ below.

The median main-sequence residual of CG_4 galaxies is:

- marginally different from that of Control4B, with a median offset of $\Delta = 0.056$ ($0.028 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.056$.
- significantly different from that of Control4C, with a median offset of $\Delta = 0.087$ ($0.067 \leq \Delta \leq 0.128$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.009$.
- not significantly different from that of RG4, with a median offset of $\Delta = 0.046$ ($-0.037 \leq \Delta \leq 0.098$, 16th–84th percentile bootstrap interval), corresponding to $p = 0.532$.

Because the raw residual comparison and the matched comparison below condition on different covariate structures, and because the apparent offset changes in strength and sign after matching, we treat SFMS residuals as a secondary diagnostic rather than as evidence for a robust compact-group star-formation offset.

Within CG_4, the star-forming main-sequence offsets of the Embedded, Isolated and Predominant classes remain statistically consistent with one another.

4. Environmental and group-scale diagnostics

4.1. Distance to the BGG

We also compare the sSFR of satellite galaxies with their distance to the BGG, expressed both in projected kpc and in units of $\langle R_{ij} \rangle$.

Significant monotonic trends are detected in the following cases.

In CG_4, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.24$ for 174 satellites ($p = 1.5 \times 10^{-3}$) and Pearson coefficient $r = 0.21$ ($p = 5.5 \times 10^{-3}$).

In $\mathrm{Control}_{4B}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.08$ for 1972 satellites ($p = 3.7 \times 10^{-4}$) and Pearson coefficient $r = 0.08$ ($p = 2.4 \times 10^{-4}$).

In $\mathrm{Control}_{4C}$, sSFR correlates with projected distance to the BGG with Spearman coefficient $\rho_S = 0.27$ for 1992 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.23$ ($p < 10^{-4}$).

In $\mathrm{Control}_{4C}$, sSFR correlates with distance to the BGG normalized by $\langle R_{ij} \rangle$ with Spearman coefficient $\rho_S = 0.10$ for 1992 satellites ($p < 10^{-4}$) and Pearson coefficient $r = 0.10$ ($p < 10^{-4}$).

The supporting distance plots are shown in Appendix Figs. C.5 and C.6.

4.2. Dominance

We split groups into dominated systems, where the BGG contributes more than 60% of the group luminosity, and non-dominated systems. This split does not produce a strong compact-group signal by itself: CG_4 shows no significant dominated/non-dominated difference after the Benjamini–Hochberg false-discovery-rate (FDR) correction. Here p_{adj} denotes the FDR-adjusted p-value, and Cliff's δ is a rank-based effect size ranging from -1 to 1.

Across the control and reference samples, the clearest binary separation is instead in group luminosity. Dominated systems have lower L_{group} than their non-dominated counterparts: $\mathrm{Control}_{4B}$ gives 9.1×10^{10} versus 1.28×10^{11} $L_\odot$ ($p_{\mathrm{adj}} < 10^{-4}$, $\delta = -0.55$); $\mathrm{Control}_{4C}$ gives 8.73×10^{10} versus 9.98×10^{10} $L_\odot$ ($p_{\mathrm{adj}} < 10^{-4}$, $\delta = -0.32$); and RG_4 gives 7.09×10^{10} versus 9.82×10^{10} $L_\odot$ ($p_{\mathrm{adj}} = 1 \times 10^{-4}$, $\delta = -0.59$). The negative δ values mean that dominated groups tend to occupy the lower-luminosity side of each comparison.

The spiral-fraction correlations show a broader environmental pattern rather than a clean dominated/non-dominated dichotomy. The most coherent set appears in $\mathrm{Control}_{4\mathrm{C}}$, where S_{frac} increases with $\left\langle R_{ij} \right\rangle$ in both dominated and non-dominated groups ($\rho_S = 0.40$, $p < 10^{-4}$ and $\rho_S = 0.38$, $p < 10^{-4}$), while weaker negative trends are found with σ_v and L_{group}. $\mathrm{Control}_{4\mathrm{B}}$ contributes only weaker non-dominated-group correlations. In CG_4, the significant trends occur in small subsets, with S_{frac} decreasing with L_{group} among dominated groups ($\rho_S = -0.50$, 36 groups) and with $\Delta V_{\mathrm{BGG}}/\sigma_v$ among non-dominated groups ($\rho_S = -0.53$, 26 groups); we therefore treat these as supporting diagnostics rather than the main dominance result. The full distributional and correlation diagnostics are shown in Appendix Figs. C.7 and C.8.

4.3. Morphology and domination

When we further split the samples by domination and by the morphology of the BGG, significant effects only appear in the following cases.

- $\mathrm{Control}_{4B}$ dominated groups, where the median satellite spiral fraction is 1.00 for spiral BGGs and 0.67 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.842 ($p = 0.01$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.842 ($p = 0.02$).
- $\mathrm{Control}_{4C}$ dominated groups, where the median satellite spiral fraction is 0.67 for spiral BGGs and 0.50 for elliptical BGGs. The association between BGG morphology and satellite spiral fraction has a group-level odds ratio of 1.918 ($p < 10^{-4}$) in this sample; the same within-sample contrast under a satellite-level clustered uncertainty treatment gives an odds ratio of 1.918 ($p < 10^{-4}$).

In each case the group-level and satellite-level clustered treatments share the same point estimate by construction and differ only in the uncertainty model, which is why the odds ratios coincide.

4.4. Morphology versus Zheng–Shen class, magnitude gap, and BGG dominance

We next test whether the E/Sp morphology mix is associated with three group-level descriptors: the Zheng–Shen compact-group class, the first–second magnitude gap $\Delta m_{12} = M_{r,2} - M_{r,1}$, and the continuous BGG luminosity fraction $f_{L,\mathrm{BGG}} = L_{\mathrm{BGG}}/L_{\mathrm{group}}$. These tests use only the existing Galaxy Zoo elliptical/smooth and spiral/features classes; uncertain morphologies are excluded from the binary fits. We run the tests separately for all galaxies, BGGs only, and satellites only, so that any trend driven by the BGG itself is not folded into the satellite result. For satellites, $f_{L,\mathrm{BGG}}$ mechanically includes the satellite luminosity in the group-luminosity denominator, so any satellite-level trend with $f_{L,\mathrm{BGG}}$ is partly definitional; we therefore report the satellite models with a stellar-mass control.

The magnitude-gap and BGG-luminosity-fraction quantities are available for all 118 CG_4+RG_4 groups used in this test. The Zheng–Shen class is available only for compact groups. Split systems are absent from the current CG_4 sample because they are removed before the paper analysis, and no corresponding Zheng–Shen class is defined for the RG_4 catalogue used here. We therefore report the RG_4 class test as unavailable, while retaining RG_4 in the Δm_{12} and $f_{L,\mathrm{BGG}}$ comparisons.

Overall, these tests provide little evidence that the E/Sp mix is driven by Zheng–Shen class, magnitude gap, or BGG luminosity fraction. The few marginal associations are therefore treated as descriptive diagnostics rather than as robust secondary trends.

For the Zheng–Shen compact-group class, no CG_4 association reaches the conventional 5% significance threshold. The closest case is the satellite-only test, which is treated as marginal and descriptive. Because the Zheng–Shen class is not defined for the RG_4 catalogue, we do not attempt an RG_4 class comparison.

For Δm_{12}, no statistically significant morphology trend is found in CG_4 or RG_4, whether the fit is run for all

Table 6. Elliptical/smooth and spiral/features counts by Zheng–Shen class for the CG_4 sample. Uncertain morphologies are excluded. Split systems have zero counts because they were removed by the existing non-split CG_4 sample definition.

Zheng–Shen class	All CG_4 E/Sp	CG_4 BGG E/Sp	CG_4 sat. E/Sp
Split	0/0	0/0	0/0
Isolated	7/11	3/1	4/10
Predominant	41/22	11/5	30/17
Embedded	76/53	22/11	54/42

Table 7. Tests of the E-vs-Sp morphology mix against compact-group class, magnitude gap, and BGG luminosity fraction. The continuous columns report the available adjusted logistic model for $P(\mathrm{E})$; all-galaxy and satellite fits use group-clustered standard errors. Odds ratios are reported per magnitude for Δm_{12} and per 0.1 increase in luminosity fraction for $f_{L,\mathrm{BGG}}$.

Subset	Sample	Test	Result
all galaxies	CG_4	Class	$p = 0.137$
		Δm_{12}	OR 1.42, $p = 0.167$
		$f_{L,\mathrm{BGG}}$	OR 1.29, $p = 0.080$
all galaxies	RG_4	Class	not defined
		Δm_{12}	OR 1.33, $p = 0.220$
		$f_{L,\mathrm{BGG}}$	OR 1.09, $p = 0.540$
BGGs only	CG_4	Class	$p = 1.000$
		Δm_{12}	OR 1.77, $p = 0.426$
		$f_{L,\mathrm{BGG}}$	OR 1.68, $p = 0.174$
BGGs only	RG_4	Class	not defined
		Δm_{12}	OR 1.58, $p = 0.312$
		$f_{L,\mathrm{BGG}}$	OR 1.21, $p = 0.520$
satellites only	CG_4	Class	$p = 0.066$
		Δm_{12}	OR 1.28, $p = 0.346$
		$f_{L,\mathrm{BGG}}$	OR 1.18, $p = 0.321$
satellites only	RG_4	Class	not defined
		Δm_{12}	OR 1.41, $p = 0.342$
		$f_{L,\mathrm{BGG}}$	OR 1.12, $p = 0.584$

galaxies, BGGs only, or satellites only. The same conclusion holds for $f_{L,\mathrm{BGG}}$: no fitted trend reaches $p < 0.05$. The CG_4–RG_4 interaction models detect no significant difference in the morphology–dominance relation for either Δm_{12} or $f_{L,\mathrm{BGG}}$, across all galaxies, BGGs only, and satellites only. A class interaction is not defined because the Zheng–Shen compact-group class has no RG_4 counterpart in the present catalogue. Across the morphology–class, morphology–gap, morphology–luminosity-fraction, and interaction tests, no result remains significant after Benjamini–Hochberg FDR control.

4.5. Crossing time

The crossing-time diagnostic mainly separates systems by dynamical scale rather than by luminosity alone. Shorter t_{cross} systems have larger virial mass-to-light ratios and virial masses: the correlations with $\log(\mathcal{M}_{\mathrm{VT}}/L_r)$ and $\log(\mathcal{M}_{\mathrm{VT}})$ are $\rho_{\mathrm{S}} = -0.39$ ($p < 10^{-4}$) and $\rho_{\mathrm{S}} = -0.33$ ($p < 10^{-4}$), respectively, over 816 groups. By contrast, the relation with total group luminosity is formally detected but very weak ($\rho_{\mathrm{S}} = 0.08$). We therefore read t_{cross} as a dynamical compactness indicator tied primarily to virial mass and mass-to-light ratio, with little independent leverage from L_{group}. Appendix Fig. C.9 shows the corresponding projected trend.

4.6. Optical colours

We derive the $(u-r)$, $(u-g)$, $(g-r)$ and $(r-i)$ colours from the matched SDSS photometry. The retained colour-analysis matches include 110 of 248 galaxies in CG_4, 931 of 2792 galaxies in $\mathrm{Control}_{4B}$, 1368 of 2816 galaxies in $\mathrm{Control}_{4C}$, 102 of 224 galaxies in RG_4. This stricter subset differs from the availability audit in Fig. C.24: the figure's colour row only requires the four derived SDSS colour columns to be present in the final catalogue, giving 217 of 248 CG_4 galaxies, whereas the colour analysis here also requires successful matched-photometry retention and the covariates used below. Stellar mass is a confounding variable in colour comparisons (Alpaslan et al. 2015); we therefore model each colour jointly with $\log(M_\star/M_\odot)$ rather than comparing unadjusted colour distributions.

The colour-analysis subset is not representative of the full galaxy catalogues: matched galaxies are systematically less massive, more actively star-forming, and more frequently satellites than unmatched galaxies. The fractions retained in this stricter colour-analysis subset are 44.4% in CG4, 33.3% in Control4B, 48.6% in Control4C, 45.5% in RG4. The colour analysis is therefore useful as a secondary diagnostic, but it cannot carry the main interpretation by itself.

The raw catalogue colour–mass fits indicate that CG_4 does not simply follow the same colour–mass relation as every comparison sample: common slopes are rejected in all four colours, with the significant directional slope differences concentrated in the bluer colours and dependent on the adopted comparison catalogue. In the satellite-only comparison, the fitted offsets and slopes likewise depend on the control definition. These tests motivate the robustness analysis rather than establishing a stand-alone compact-group colour effect.

We therefore refit the satellite colours with cluster-robust standard errors, using catalogue-qualified group identifiers. In the pooled comparison adjusted only for stellar mass and redshift, none of the four CG_4 coefficients is significantly negative after Holm correction. Adding morphology and sSFR class does not reveal a significant negative coefficient. However, these adjusted coefficients are control-sample dependent, and they do not persist within the sSFR or morphology splits. The data therefore do not localize the colour effect to the star-forming spiral population expected for a simple interaction-triggered star-formation scenario.

Residual colour trends with projected BGG-centric distance are consistent with stronger processing near the group centre and a less processed outer population, but these trends use the same biased colour subset and are treated as exploratory. The group-level BGG–satellite colour concordance is similarly secondary: redder BGGs tend to have redder mean satellite populations in the combined group sample, but the CG_4 slope is not a robustly distinct population-level signature. Detailed colour–mass, satellite, and radial diagnostics are shown in Appendix Figs. C.10–C.12. Figure 4 summarizes the model and control-sample dependence that motivates our interpretation: the present data do not establish a robust global blue or red offset for CG_4 satellites.

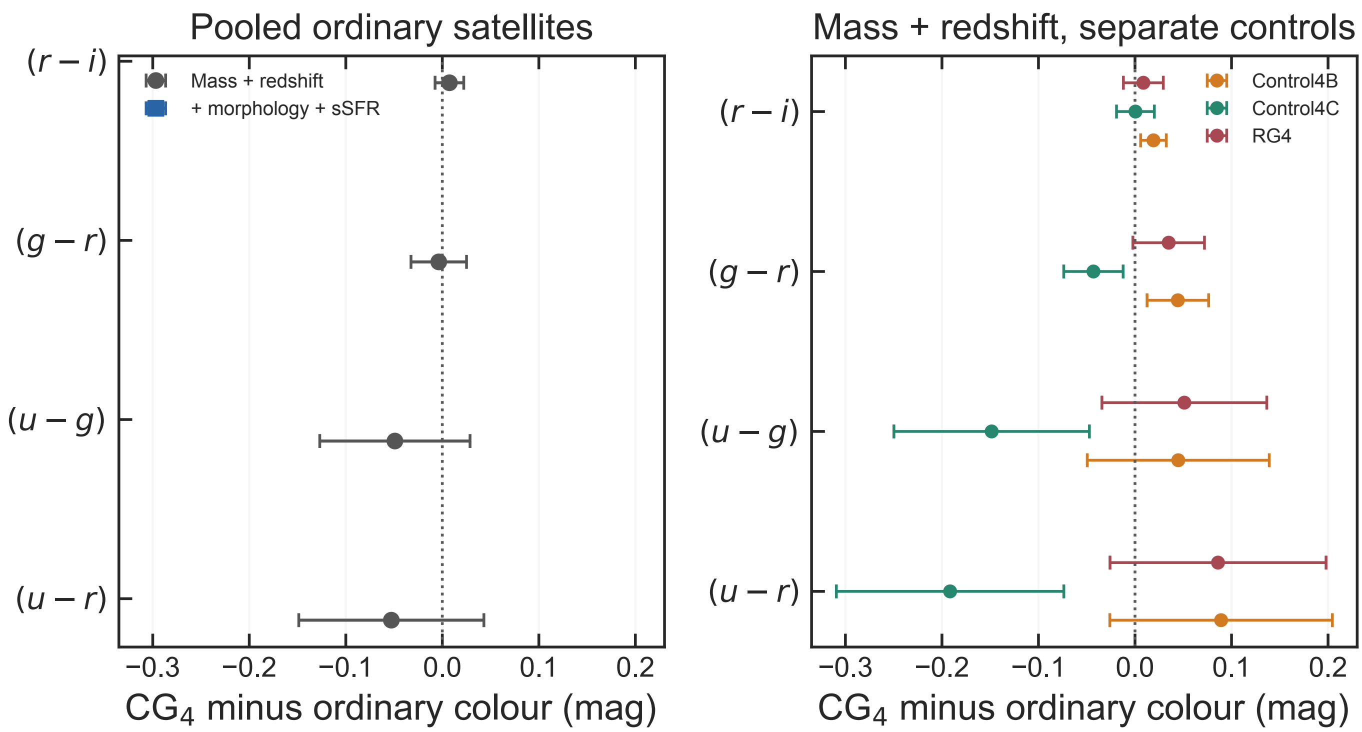

Fig. 4. Robustness of the satellite colour coefficients. Left: pooled ordinary-group comparison before and after controlling for morphology and sSFR class. Right: mass- and redshift-adjusted comparisons against each control catalogue separately. Points show the CG$_4$ minus ordinary coefficient and bars show 95% confidence intervals; negative values indicate bluer CG$_4$ satellites.

5. Robustness: what drives the compact-group differences?

The previous sections establish that CG$_4$ galaxies differ from the control catalogues in their population mix, especially in morphology and in the star-forming fraction of satellites. We now ask whether these differences survive after controlling for stellar mass, rank and available group-scale quantities, and whether they are better interpreted as compact-group-specific evolution, preprocessing in dense halo cores, or catalogue-selection effects.

5.1. Mass-, rank- and environment-adjusted galaxy properties

We fit binomial models to the combined galaxy catalogues, standardizing the continuous predictors and using group-clustered standard errors. The common adjustment set includes stellar mass, redshift, satellite/BGG rank, group luminosity and velocity dispersion when sufficiently complete.

For quenched status, the CG$_4$ odds ratio is 1.08, with 95% confidence interval $[0.80, 1.44]$ and Holm-corrected $p = 1.000$.

For quenched satellite status, the CG$_4$ odds ratio is 1.17, with 95% confidence interval $[0.86, 1.58]$ and Holm-corrected $p = 1.000$.

For star-forming satellite status, the CG$_4$ odds ratio is 0.86, with 95% confidence interval $[0.63, 1.16]$ and Holm-corrected $p = 1.000$.

For elliptical morphology, the CG$_4$ odds ratio is 1.50, with 95% confidence interval $[1.13, 1.99]$ and Holm-corrected $p = 0.032$.

For elliptical satellite morphology, the CG$_4$ odds ratio is 1.71, with 95% confidence interval $[1.24, 2.34]$ and Holm-corrected $p = 0.007$.

For elliptical BGG morphology, the CG$_4$ odds ratio is 0.77, with 95% confidence interval $[0.39, 1.53]$ and Holm-corrected $p = 1.000$.

For spiral morphology, the CG$_4$ odds ratio is 0.67, with 95% confidence interval $[0.50, 0.89]$ and Holm-corrected $p = 0.032$.

For spiral satellite morphology, the CG$_4$ odds ratio is 0.59, with 95% confidence interval $[0.43, 0.80]$ and Holm-corrected $p = 0.007$.

Compact-group membership therefore remains independently associated with morphology after the available adjustments: CG$_4$ galaxies have a higher probability of being elliptical and a lower probability of being spiral. In contrast, the adjusted quenched and star-forming satellite coefficients are weaker and do not provide evidence for an instantaneous star-formation difference that remains significant after correction.

The adjusted morphology signal is primarily carried by satellites rather than BGGs: the satellite-only models give a smooth/elliptical odds ratio of 1.71 and a spiral/features odds ratio of 0.59, whereas the BGG smooth/elliptical coefficient is not significant after adjustment (OR 0.77; Holm-corrected $p = 1.000$).

5.2. Morphology robustness to projected crowding

Because Galaxy Zoo morphology is image-based, the CG$_4$ smooth/elliptical excess must be checked against projected crowding and deblending. We therefore repeat the morphology comparison after excluding galaxies whose nearest

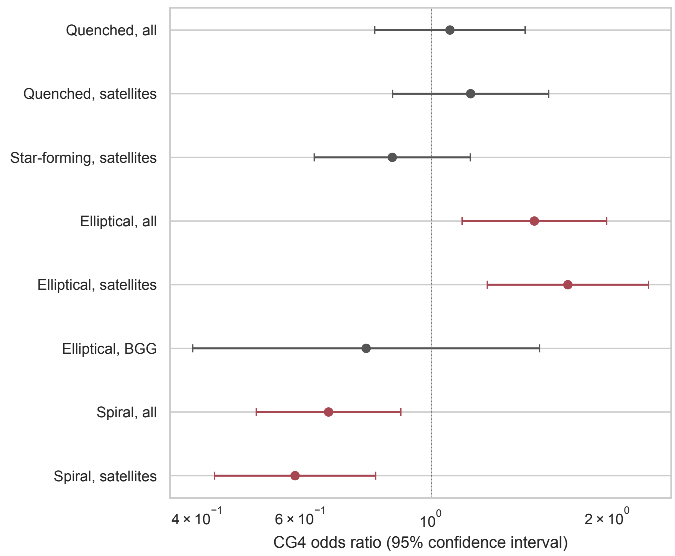

Fig. 5. Adjusted CG_4 odds ratios for star-formation and morphology diagnostics, including all-galaxy, satellite-only, and BGG-only morphology rows. Bars show 95% confidence intervals; the vertical line marks equal odds.

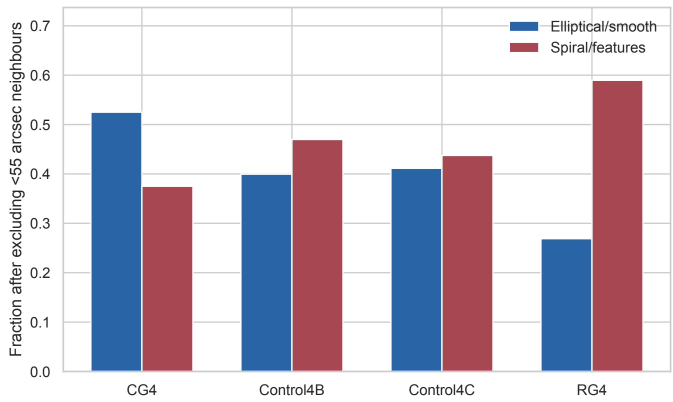

Fig. 6. Galaxy Zoo morphology fractions after excluding galaxies with a nearest projected neighbour within 55 arcsec. The CG_4 smooth/elliptical excess persists after this closest-neighbour exclusion.

projected neighbour within the catalogue group lies within 55 arcsec. The retained samples contain 200 CG_4 galaxies and 2455 $Control_{4C}$ galaxies. After this exclusion, the CG_4 smooth/elliptical fraction is 52.5%, compared with 41.1% in $Control_{4C}$; the CG_4–$Control_{4C}$ exact-test result remains significant ($p = 0.002$). The CG_4 smooth fraction rises relative to the raw full-sample value in Table 2, rather than falling after the closest projected pairs are removed. This argues against the simplest interpretation that the CG_4 smooth excess is produced solely by crowding in the closest pairs.

The complementary close-neighbour-flag model reaches the same qualitative conclusion. After adjusting for the standard covariates and a < 55-arcsec neighbour flag, the CG_4 smooth/elliptical odds ratio is 1.52 (95% CI $[1.13, 2.04]$; $p = 0.006$). The close-neighbour flag is itself associated with smooth morphology (OR 2.07; $p < 10^{-4}$), so this test does not prove that all Galaxy Zoo classifications are immune to crowding. It does rule out the simplest closest-pair artefact as the sole origin of the morphology result. Figure 6 shows the corresponding post-exclusion morphology fractions.

5.3. Matched-control comparison

We matched 234 of the 248 available CG_4 galaxies one-to-one to 234 ordinary-group galaxies without replacement, requiring common propensity-score support, a 0.2-standard-deviation caliper and exact rank strata. The remaining 14 CG_4 galaxies were excluded because they lay outside the common propensity-score support or exceeded the adopted caliper. The propensity model uses $\log M_\star$, redshift, rank, $\log L_{\rm group}$, and σ_v. Projected distance is retained as a phase-space and interaction diagnostic rather than matched away because it partly defines compactness and has limited overlap. The largest absolute standardized mean difference changed from 0.24 before matching to 0.11 afterwards. The residual imbalance is carried by the group-luminosity covariate ($\log L_{\rm group}$); it lies slightly above the conventional 0.1 threshold, which is why we read the matched contrasts together with the adjusted models rather than in isolation.

The matched CG_4–minus–control difference in quenched fraction is −0.022, with 95% bootstrap interval $[-0.100, 0.055]$ and Holm-corrected $p = 1.000$.

After matching, this does not provide evidence for an enhanced quenched fraction in CG_4 that remains significant after correction.

The corresponding CG_4–minus–control difference in star-forming fraction is 0.022, with 95% bootstrap interval $[-0.055, 0.100]$ and Holm-corrected $p = 1.000$.

The matched CG_4–minus–control difference in elliptical fraction is 0.087, with 95% bootstrap interval $[-0.005, 0.179]$ and Holm-corrected $p = 0.361$.

The matched CG_4–minus–control difference in spiral fraction is −0.087, with 95% bootstrap interval $[-0.179, 0.005]$ and Holm-corrected $p = 0.361$.

Among matched star-forming galaxies, the residual-sSFR difference is 0.074, with 95% bootstrap interval $[-0.075, 0.226]$ and $p = 0.332$.

The matched-outcome Holm correction is intentionally conservative: quenched/star-forming and elliptical/spiral diagnostics are retained in the same family even though the audit finds that they are exact complements on their complete-case subsets. Using a non-redundant family would not weaken the morphology conclusion.

The main interpretation does not rely on the exact GMM division of the sSFR plane. Repeating the key satellite comparison with a fixed threshold, $\log_{10}(\mathrm{sSFR/yr^{-1}}) = -11.0$, gives CG_4 and RG_4 satellite star-forming fractions of 41.4% and 60.2%, respectively, a CG_4–RG_4 difference of −18.9 percentage points ($p = 5.3 \times 10^{-4}$). The qualitative hierarchy is unchanged: the morphology result is independent of the sSFR threshold, whereas the quenched/star-forming contrast remains more sensitive to modelling choices.

The matched comparison is therefore consistent with the adjusted models: the most robust compact-group signal is morphological, while the quenched and star-forming-fraction signals are more model-dependent.

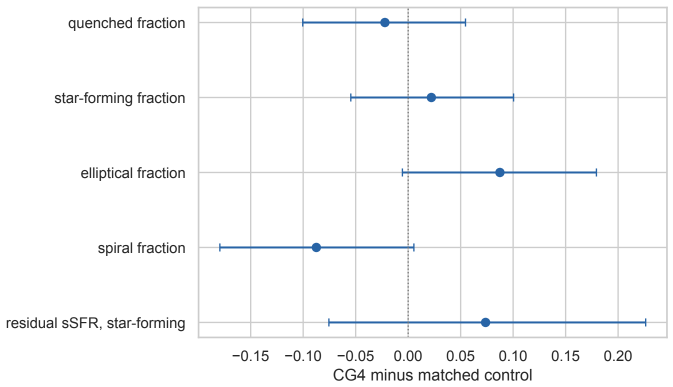

Fig. 7. CG_4–minus–control effects after one-to-one matching. Bars show 95% bootstrap confidence intervals.

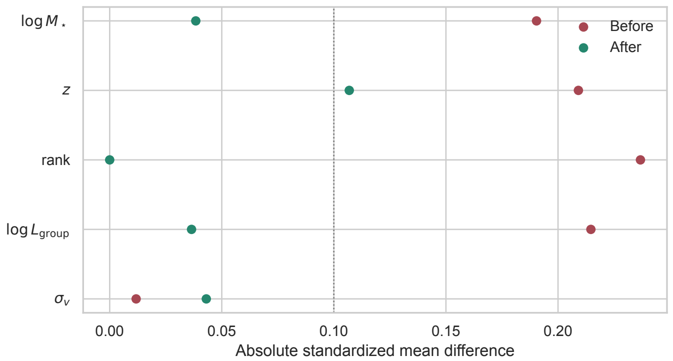

Fig. 8. Absolute standardized mean differences before and after matching.

5.4. Satellite segregation around the brightest group galaxy

The previous sections show global differences between CG_4 and RG_4 satellites. We now ask whether these differences are concentrated near the BGG, as expected for accelerated evolution in the dense inner group, or whether they persist across the satellite population. The analysis uses 174 CG_4 and 166 RG_4 satellites, excludes rank-1 galaxies, ranks satellites by projected BGG-centric distance, and measures quenched and early-type fractions with group-cluster bootstrap uncertainties. When available, we also use $|\Delta v_{\rm BGG}|/\sigma_v$, with $\Delta v_{\rm BGG} = c(z_{\rm gal} - z_{\rm BGG})/(1 + z_{\rm group})$, as a projected phase-space diagnostic. Similar projected phase-space coordinates are often used as observational proxies for satellite infall state or quenching time, but only statistically (Mahajan et al. 2011; Oman & Hudson 2016; Wetzel et al. 2013).

In the innermost satellite-distance bin, the quenched fractions are 68.3% for CG_4 and 48.2% for RG_4, a CG_4–RG_4 difference of 20.1 percentage points. The corresponding early-type fractions are 64.7% and 35.6%, with a difference of 29.2 percentage points.

The quenched-fraction contrast is similar between the inner and outer distance ranks, so the data do not isolate the CG_4 offset to the immediate BGG environment.

After adjusting for stellar mass, redshift, distance bin, and available group-scale controls, the quenched-status model gives a CG_4 odds ratio of 1.18 (95% CI 0.43–3.24; $p = 0.742$).

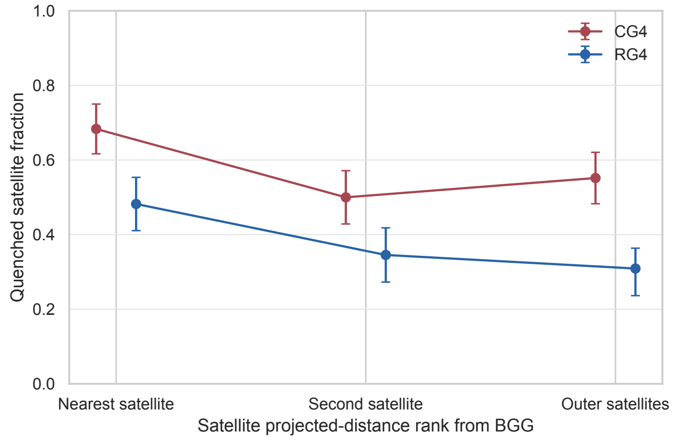

Fig. 9. Quenched satellite fraction as a function of projected-distance rank from the BGG. Points show CG_4 and RG_4 fractions; bars show group-cluster bootstrap 16th–84th percentile intervals.

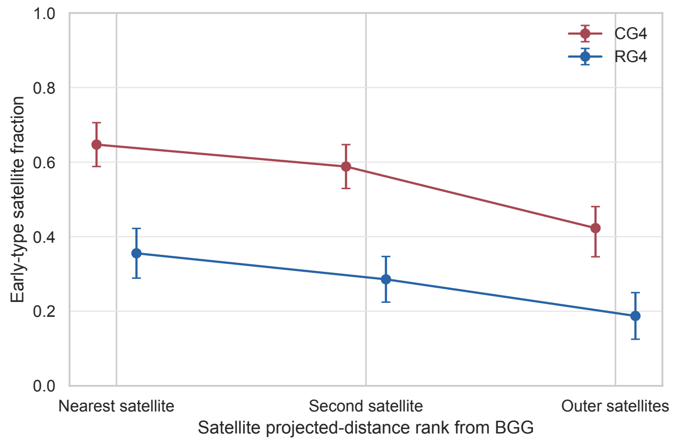

Fig. 10. Early-type satellite fraction as a function of projected-distance rank from the BGG. Because the catalogue separates ellipticals, spirals, and uncertain morphologies but not S0 galaxies, early type is equivalent to elliptical in this diagnostic.

The analogous early-type model gives a CG_4 odds ratio of 2.42 ($p = 0.078$).

A nearest-neighbour mass–redshift matched check uses 126 CG_4–RG_4 satellite pairs. The matched quenched-fraction difference is 9.5 percentage points, and the matched early-type difference is 18.7 percentage points.

Projected phase-space coordinates should not be read as orbital reconstructions for individual galaxies. In compact groups the member counts are small, velocity dispersions are noisy, and projection mixes recently accreted and long-resident satellites. We therefore treat these bins as an observational diagnostic of where the CG_4–RG_4 population contrast appears, rather than as a direct dynamical clock.

5.5. Magnitude gaps and fossil-like assembly

We define $\Delta m_{12} = M_{r,2} - M_{r,1}$ and $\Delta m_{14} = M_{r,4} - M_{r,1}$, so that a more dominant BGG has a positive, larger gap.

For Δm_{12}, the CG_4 and control medians are 1.17 and 1.10 mag, respectively (Holm-corrected $p = 0.824$).

For Δm_{14}, the CG_4 and control medians are 2.56 and 2.59 mag, respectively (Holm-corrected $p = 0.824$).

The compact-group morphology signal is therefore not accompanied by a significant excess in these simple fossil-like magnitude-gap indicators.

The correlation between Δm_{12} and quenched satellite fraction is $\rho_S = -0.14$ (Holm-corrected $p < 10^{-4}$).

The supporting magnitude-gap distributions are shown in Appendix Figs. C.20 and C.21.

5.6. Recently quenched candidates

The catalogues do not contain both D_n4000 and $H\delta_A$, so a post-starburst classification is not possible. We therefore perform only a limited Hα-equivalent-width comparison, using the SDSS convention in which negative equivalent width denotes emission. We classify strong Hα emission using Hα EW ≤ -3 Å.

The strong-emission fraction is 32.3% in CG$_4$, 41.0% in Control$_{4B}$, 44.2% in Control$_{4C}$, 56.9% in RG$_4$.

Relative to Control$_{4B}$, the CG$_4$ strong-emission fraction differs by -0.087 (Holm-corrected $p = 0.012$).

Relative to Control$_{4C}$, the CG$_4$ strong-emission fraction differs by -0.119 (Holm-corrected $p = 0.001$).

Relative to RG$_4$, the CG$_4$ strong-emission fraction differs by -0.246 (Holm-corrected $p < 10^{-4}$).

These limited Hα results indicate that CG$_4$ galaxies have a lower strong-emission fraction than in each control sample in the current catalogue, but they do not provide a direct post-starburst or recently-quenched classification.

The Hα equivalent-width distributions are shown in Appendix Fig. C.22.

5.7. Emission-line and AGN-like populations

The complementary emission-line analysis classifies BPT star-forming, composite, and AGN-like populations for galaxies with the required diagnostic line ratios: composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations in the [N II] BPT plane, and AGN-like galaxies lie above the Kewley et al. (2001) line; galaxies with a pre-existing AGN flag but no BPT line ratios are also counted as AGN-like. BPT-unclassified galaxies are excluded from the AGN-like fraction denominator rather than treated as confirmed non-AGN. Among BPT-classified CG$_4$ galaxies, the AGN-like fraction is 38.0%.

After mass and environment adjustment, the CG$_4$ AGN-like odds ratio is 1.14 ($p = 0.447$).

Thus, AGN-like activity does not provide a robust independent compact-group discriminator in this analysis.

The AGN-like fractions are shown in Appendix Fig. C.23.

5.8. Pairwise tidal index

The pairwise analysis reconstructs projected separations and velocity differences for 6080 galaxies and refits the adjusted models with and without the pairwise tidal index $\sum_j M_{\star,j}/R_{ij}^3$ (in $M_\odot\,\mathrm{kpc}^{-3}$). The two model variants estimate *different quantities*, and we state both estimands explicitly. The model without the index estimates the *total* association between compact-group membership and the outcome at fixed stellar mass, redshift, rank, and group-scale covariates. The model with the index estimates the association *at fixed projected local density*. Because the index is built from R^{-3}-weighted projected separations, it is mathematically entangled with the compact-group selection itself (compact groups are selected to have small projected separations); conditioning on it therefore partially conditions on the exposure. A reduction of the CG$_4$ coefficient when the index is added is thus *conditional attenuation*, not mediation: it does not show that local density accounts for the morphology contrast, and it cannot, because the causal ordering of selection, density, and morphology is not identified by projected snapshots. The baseline models in this subsection are refit on the pairwise-analysis complete-case sample, so their coefficients need not be numerically identical to the corresponding coefficients in Sect. 5.1. Because R_{ij} is projected, the term is a local projected-density and immediate-neighbour proxy rather than a direct tidal-history measurement.

For quenched status, the baseline CG$_4$ odds ratio is 1.17 ($p = 0.291$), while the model including the tidal index gives a CG$_4$ odds ratio of 0.77 ($p = 0.106$). The tidal-index term itself has odds ratio 1.68 ($p < 10^{-4}$).

For elliptical morphology, the baseline CG$_4$ odds ratio is 1.64 ($p = 4.4 \times 10^{-4}$), whereas the model including the tidal index gives a CG$_4$ odds ratio of 1.13 ($p = 0.421$). This baseline is the pairwise complete-case refit, not the full adjusted model in Sect. 5.1. The tidal-index term has odds ratio 1.58 ($p < 10^{-4}$). The projected tidal-density proxy therefore *attenuates* the CG$_4$ coefficient, indicating that the morphology contrast is concentrated in the most extreme local projected configurations; given the selection entanglement described above, this conditional attenuation must not be read as the index accounting for the effect, and the result remains an exploratory projected-pair diagnostic.

5.9. Galaxy sizes at fixed stellar mass

We now test whether CG$_4$ galaxies differ in size at fixed stellar mass, using the seeing-corrected Simard half-light radii as the primary measure and the Petrosian radii as the model-independent cross-check (Sect. 2.4). The Simard measure resolves 77.4% of the CG$_4$ galaxies and 72.2% of the pooled control galaxies, while the Petrosian radius is essentially complete in every sample (Appendix Fig. C.13). Because the CG$_4$–control difference in Simard completeness exceeds five percentage points, every Simard-based coefficient below carries a differential-completeness caveat and is read together with the Petrosian rerun, which does not share this incompleteness. The descriptive mass–size relations and the per-sample sizes predicted at $\log(M_\star/M_\odot) = 10.3$ are shown in Appendix Fig. C.14.

For satellite galaxy size, the CG$_4$ offset at fixed stellar mass, redshift, and available group-scale covariates is -0.000 dex (-0.1%), with 95% confidence interval $[-0.030, 0.030]$ and Holm-corrected $p = 1.000$, over 3878 satellites in 1482 groups.

The all-galaxy model gives 0.000 dex (95% CI $[-0.026, 0.027]$; Holm-corrected $p = 1.000$), and the BGG-only model gives 0.008 dex (Holm-corrected $p = 1.000$).

No coefficient in this primary family remains significant after Holm correction, so the pooled adjusted models do not establish a CG$_4$ size offset at fixed stellar mass.

The per-control versions of the mass- and redshift-adjusted fit are descriptive and shown in Appendix Fig. C.15; the fitted CG$_4$ coefficient is -0.035 dex against

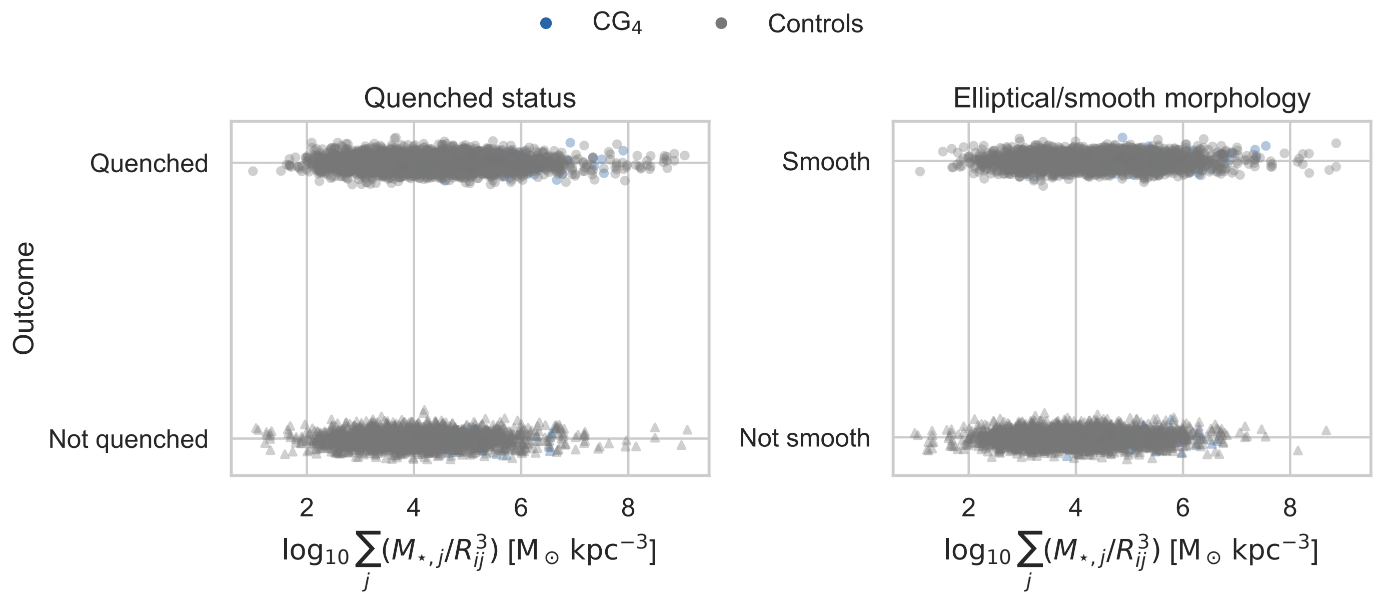

Fig. 11. Quenched status and elliptical/smooth morphology as functions of the reconstructed pairwise tidal index. Colours distinguish CG_4 from control galaxies.

$Control_{4B}$, 0.027 dex against $Control_{4C}$, −0.006 dex against RG_4 (raw p-values with Benjamini–Hochberg control inside this figure family). The sign and significance of the CG_4 coefficient depend on the adopted control, so we do not interpret any single per-control contrast on its own.

The one-to-one matched comparison re-derives the same 234 propensity pairs as the matched-control analysis, with a separate Holm family for the three size outcomes so that the published matched-outcome family is untouched. On the 160 pairs in which both members have valid Simard sizes, the mean CG_4–minus–control difference in $\log_{10} R_{\rm chl,r}$ is 0.027 dex (95% bootstrap interval $[-0.013, 0.065]$; Holm-corrected $p = 0.556$). The corresponding Petrosian difference is 0.005 dex (Holm-corrected $p = 1.000$), and the matched concentration difference is 0.005 (Holm-corrected $p = 1.000$).

The crowding checks mirror Sect. 5.2: excluding galaxies with a projected neighbour within 55 arcsec leaves a satellite coefficient of −0.001 dex ($p = 0.942$), and adding the close-neighbour flag as a covariate gives 0.001 dex ($p = 0.972$).

The Petrosian rerun, with the field seeing as an additional standardized covariate, gives a satellite coefficient of −0.015 dex (Holm-corrected $p = 0.413$) and an all-galaxy coefficient of −0.013 dex (Holm-corrected $p = 0.413$).

The per-galaxy difference between the two measures, $d = \log R_{\rm chl,r} - \log R_{50,r}$, averages 0.048 dex, and is larger for elliptical/smooth galaxies (0.068 dex) than for spiral/features galaxies (0.038 dex), as expected if the Petrosian aperture preferentially truncates concentrated profiles. Its rank correlation with the nearest-neighbour separation is $\rho_S = -0.000$ ($p = 0.984$), so residual blending does not drive a strong measure-dependent size systematic (Appendix Fig. C.16).

Adding the pairwise projected tidal index of Sect. 5.8 to the all-galaxy size model changes the CG_4 coefficient from 0.000 dex to 0.027 dex on the pairwise complete-case sample; with a near-null baseline coefficient, the attenuation diagnostic used for morphology is not informative here.

The radial diagnostic is exploratory: mass- and redshift-adjusted satellite size residuals correlate with projected BGG-centric distance with $\rho_S = 0.18$ ($p = 0.023$) in CG_4 and $\rho_S = 0.19$ ($p < 10^{-4}$) in the pooled controls (Appendix Fig. C.18). A more negative inner-region residual would be the tidal-truncation expectation; the observed trends are similar in CG_4 and the controls, so the data do not isolate a CG_4-specific inner-region size suppression.

The concentration index $C = R_{90,r}/R_{50,r}$ provides a model-independent structural proxy: the adjusted satellite coefficient is 0.065 (Holm-corrected $p = 0.413$), so CG_4 membership does not significantly shift the concentration index at fixed mass.

Within morphology classes, the satellite-only fits give 0.035 dex for elliptical/smooth satellites (Holm-corrected $p = 0.204$) and 0.000 dex for spiral/features satellites (Holm-corrected $p = 0.984$), and the CG_4×morphology interaction is 0.035 dex ($p = 0.299$). These strata are secondary diagnostics; they test whether any size offset is independent information beyond the morphology signal rather than a restatement of it.

5.10. Selection diagnostics

The fractions with complete measurements for the major derived quantities are shown in Appendix Fig. C.24. The colour-matched versus unmatched comparisons quantify shifts in stellar mass, redshift, rank, star-formation class and morphology.

The median nearest-neighbour angular separation is 100.9 arcsec in CG_4 and 210.5 arcsec in the controls; the fractions below the 55-arcsec SDSS fibre-collision scale are 19.4% and 9.2%, respectively.

Under the availability definitions shown in Fig. C.24, no major availability difference meets both criteria of corrected significance and a difference exceeding 10 percentage points.

The group-scale availability row is based on the group covariates available for this catalogue: radius scale, velocity dispersion, group luminosity, and BGG luminosity dominance. A harmonized group-mass estimate is not available in these tables, so models that request it omit that covariate.

The colour-selection mass audit is shown in Appendix Fig. C.25.

6. Discussion: drivers of structural evolution in compact groups

6.1. Local projected tidal density and morphology signal

The most robust residual difference between CG_4 and the regular-group controls is morphological. CG_4 galaxies are more likely to be classified as elliptical/smooth and less likely to be classified as spiral/features, and this result persists in both the adjusted binomial models and the one-to-one matched comparison. The result hierarchy therefore points to accumulated structural transformation more strongly than to a present-day star-formation offset.

The pairwise tidal-index experiment strengthens this interpretation while also narrowing it. For elliptical/smooth morphology, the pairwise complete-case baseline CG_4 odds ratio is 1.64 ($p = 4.4 \times 10^{-4}$), so it is not expected to match exactly the full adjusted-model odds ratio in Sect. 5.1. After adding the projected pairwise tidal-index proxy, the CG_4 odds ratio decreases to 1.13 ($p = 0.421$), while the tidal-index term itself has odds ratio 1.58 ($p < 10^{-4}$). This behaviour is expected if compact groups are selecting galaxies in locally dense projected configurations where repeated close companions, rather than the catalogue label alone, are linked to the smooth/early-type-like population.

This proxy uses projected separations, so it may be inflated by line-of-sight alignments and is partly entangled with compact-group selection itself: compact groups are selected to be compact in projection. We therefore read the tidal-index result as a continuous local projected-density or immediate-neighbour proxy, not as a direct time-dependent tidal-field measurement. The $\mathrm{Control}_{4C}$ comparison makes the test deliberately stringent, because $\mathrm{Control}_{4C}$ is also selected as a projected compact core around a BGG. The persistence of a CG_4–$\mathrm{Control}_{4C}$ morphology contrast in the closest-neighbour exclusion test shows that the CG_4 label is not reduced to the simplest crowding artefact. At the same time, the tidal-index model indicates that a more continuous projected-density proxy absorbs much of the CG_4 coefficient. We therefore interpret the residual CG_4–$\mathrm{Control}_{4C}$ contrast as consistent with CG_4 selecting a more extreme part of the local-density distribution, rather than as proof that compact groups define a separate causal channel.

6.2. Galaxy sizes and comparison with earlier compact-group size studies

Coenda et al. (2012) compared galaxies in the McConnachie et al. (2009) compact groups with loose-group and field samples using SDSS Petrosian half-light radii and found compact-group galaxies to be smaller at fixed luminosity, with the clearest offset for early types relative to the field. Their assessment rested on approximately 1σ comparisons of fitted size–luminosity relations, without a formal confidence level on the size difference itself. The present analysis re-poses that question with a different control design – the non-split HMCG-based CG_4 sample against three regular-group controls, including the projected-compact-core analogue $\mathrm{Control}_{4C}$ – and with seeing-corrected model sizes, a Petrosian cross-check, cluster-robust adjusted models, one-to-one matching, crowding tests, and Holm-corrected confidence intervals.

Under this stricter design, we find no compact-group size offset at fixed stellar mass relative to regular groups. This does not contradict Coenda et al. (2012) directly, because their comparison was against loose groups and the field, whereas ours is against regular-group cores selected from the same parent catalogues.

Three interpretation guardrails apply. A negative Δ at fixed mass can reflect genuine tidal truncation or stripping, a higher bulge dominance in the CG_4 population, or residual measurement blending in crowded fields. The morphology-stratified fits address the second possibility, the Petrosian cross-check and the measure-difference diagnostic address the third, and the crowding exclusions bound the sensitivity of all size coefficients to the closest projected pairs. Any future claim of compact-group size evolution should be read through those three blocks together rather than through a single pooled coefficient.

6.3. Smooth morphologies, S0 degeneracy, and transition galaxies

The Galaxy Zoo morphology variables used here separate smooth, featureless systems from galaxies with visible spiral structure or other features. They do not cleanly distinguish classical ellipticals from S0 galaxies, faded disks, or other smooth disk-dominated systems. The enhanced CG_4 elliptical/smooth fraction should therefore be interpreted as a shift toward smooth or early-type-like appearances, not as proof that compact groups specifically build classical elliptical galaxies.

This caveat is important when comparing with the recent S-PLUS structural studies of compact-group galaxies by Montaguth et al. (2023, 2025, 2026a,b). Those studies identify transition galaxies and trends in the R_e–Sérsic-index and mass-size planes, with especially strong signals in non-isolated compact groups and host structures. Their transition population can plausibly overlap with the smooth class in our catalogue: systems with faded disks, increasing central concentration, or S0-like structure may be counted as smooth here even if they are not classical ellipticals. The two approaches are therefore complementary. Their structural measurements describe the likely physical form of the transformation, while our regular-group controls show that the strongest residual compact-group signal is the accumulated smooth/featured morphology balance. With the Simard structural parameters now attached to this sample, the R_e–Sérsic-n plane is measured directly for CG_4 and its controls (Appendix Fig. C.17), turning this comparison with the S-PLUS transition-galaxy studies from a qualitative statement into a direct one.

6.4. Decoupling between morphology and instantaneous star formation

The star-formation diagnostics do not follow the morphology signal one-to-one. CG_4 satellites have a lower raw star-

Table 8. Result hierarchy. Baseline tests describe where CG_4 differs from the controls; the main interpretation is based on diagnostics that remain after adjusted or matched comparisons.

Diagnostic	Baseline CG_4 signal	Survives adjusted/-matched analysis?	Interpretation
Elliptical/smooth fraction	Enhanced in CG_4	Yes	Robust accumulated morphology signal.
Spiral/features fraction	Reduced in CG_4	Yes	Reduced late-type vote-fraction population.
Galaxy Zoo crowding check	Smooth excess persists after close-neighbour exclusion	Yes, as a limited robustness check	Morphology result not solely produced by the closest projected pairs.
Satellite star-forming fraction	Reduced, especially versus RG_4	Weak or borderline	Partly explained by mass, rank, environment, or preprocessing.
Star-forming main-sequence residual	Weak and control-dependent	No robust offset	No strong instantaneous SFR enhancement or suppression among remaining star-forming galaxies.
Satellite colours	Colour-slope and residual hints	Control-sensitive	Exploratory; not a simple uniform reddening or blueing of CG_4 satellites.
AGN-like fraction	No robust independent signal	No	Not an AGN-driven explanation.
Magnitude gaps	No significant excess	No	Not fossil-like under the simple gap diagnostics used here.
Pairwise tidal index	Local projected density proxy is informative	Reframes much of the morphology coefficient	Local projected density is linked to the morphology signal.
Galaxy size at fixed mass	Hints of smaller CG_4 sizes	No	No compact-group size effect at fixed stellar mass.
Concentration index	$R_{90,r}/R_{50,r}$ structural proxy now measured	No	No independent concentration signal at fixed mass.

forming fraction, especially relative to RG_4, but the adjusted and matched evidence is weaker than for morphology. Among galaxies that remain star-forming, the main-sequence residuals do not show a robust compact-group offset. The colour analysis is also control-sensitive and based on a biased colour-matched subset, while AGN-like fractions, magnitude gaps, and the limited Hα diagnostic do not identify a simple alternative driver.

The star-formation non-detections are also limited by the spectroscopy. CG_4 has a higher fraction of galaxies with a nearest projected neighbour below the 55-arcsec SDSS fibre-collision scale than the controls (19.4% versus 9.2%). The control value is pooled over the regular-control comparison; the $Control_{4C}$-only exclusion fraction in Sect. 5.2 is larger (12.8%) because $Control_{4C}$ is selected as the compact-core analogue. Spectra are therefore preferentially incomplete in the closest projected pairs, and short-lived interaction-triggered starbursts or post-starburst phases could be underrepresented. The MPA-JHU aperture corrections use photometric information outside the fibre (Brinchmann et al. 2004; Kauffmann et al. 2003a); these corrections may be less secure for interacting or blended compact-group members. We therefore treat the sSFR, SFMS, and Hα results as conservative secondary non-detections, not as proof that compact groups have no star-formation effect.

This decoupling suggests that the dominant observable difference is not an instantaneous quenching event caught at the present epoch. A galaxy can lose visible spiral structure, build a smoother disk or bulge-dominated profile, or fade into an S0-like state on timescales that do not require a simultaneously large sSFR residual in the surviving star-forming subset. The present data therefore favour a cumulative morphology or structure signal over a single ongoing star-formation process.

6.5. Compact groups as dense cores and preprocessing sites

The comparison samples were designed to ask whether compact groups remain special relative to ordinary group cores, not merely relative to the field. The answer is mixed but physically useful. CG_4 galaxies retain a robust morphology difference, yet much of the interpretation points toward local density, dense cores, and preprocessing rather than toward an isolated compact-group channel. This is consistent with the construction result that many compact groups are embedded in or associated with larger structures, and with the stronger structural signatures reported for non-isolated compact groups in recent S-PLUS work. We note that no harmonised halo-mass estimate is available for these catalogues (Sect. 5.10), so, although the adjusted models control for group luminosity and velocity dispersion, a residual halo-mass contribution to the morphology signal cannot be fully excluded.

The most conservative interpretation is that compact groups identify the high-density, high-interaction tail of group environments. Some systems may be genuinely compact and dynamically evolved, while others may be dense cores or substructures inside larger haloes. In both cases, the relevant physical quantity is likely the accumulated local interaction history. Future simulations and direct structural measurements, including R_e, Sérsic index, transition-galaxy classifications, and mass-size residuals, are needed to test whether the signal isolated here is unique to compact configurations or is the high-density limit of more general group preprocessing.

7. Conclusion

We have tested whether galaxies in CG_4 compact groups remain special after removing split systems and comparing them with three regular-group analogues: RG_4, $Control_{4B}$, and $Control_{4C}$. The comparison was designed to separate compact configurations from ordinary four-member groups, luminous group cores, and projected compact cores around BGGs.

The main result is morphological. CG_4 galaxies have an enhanced elliptical/smooth Galaxy Zoo fraction and a reduced spiral/features fraction. This signal survives the mass-, redshift-, rank-, and group-adjusted binomial models and is also recovered in the one-to-one matched-control analysis. In the adjusted models, the morphology signal is primarily carried by satellites; BGG morphology is not significantly different after the same adjustment.

The star-formation result is weaker. CG_4 satellites have a lower raw star-forming fraction, especially relative to RG_4, but the adjusted and matched evidence is weaker or borderline. Among galaxies that remain star-forming, CG_4 galaxies do not show a robust offset from the fitted star-forming main sequence.

The galaxy-size test adds a structural dimension to this comparison. At fixed stellar mass, redshift, and group-scale covariates, we find no significant CG_4 offset in seeing-corrected half-light radius, in either the adjusted models or the matched pairs.

The secondary diagnostics do not identify a simple alternative driver. The colour results are control-sensitive and based on a biased colour-matched subset. AGN-like fractions and magnitude gaps do not explain the morphology signal, the Hα diagnostic is limited by missing age-sensitive spectral indices, and projected phase-space trends do not localize the quenched difference uniquely to the nearest satellites. The Galaxy Zoo crowding check shows that the smooth/elliptical excess persists after excluding the closest projected pairs, so the result is not solely a closest-pair classification artefact.

We therefore interpret CG_4 galaxies as a more morphologically evolved population, consistent with accumulated transformation in very dense environments, dense group cores, and/or preprocessing. The pairwise tidal-index experiment shows that adding a local projected-density proxy reduces the compact-group morphology coefficient, so the result should not be read as evidence for a uniquely isolated compact-group quenching mechanism.

This paper now connects the regular-group control framework directly to structural measurements – R_e, Sérsic index, and mass-size residuals through the Simard catalogue – so the remaining structural future work concerns transition-galaxy classifications and better spectroscopic post-starburst diagnostics. The concentration-index check based on $R_{90,r}/R_{50,r}$ is likewise measured here: the adjusted satellite coefficient is 0.065 (Holm-corrected $p = 0.413$), so this independent structural proxy shows no significant CG_4 offset at fixed mass. Those remaining measurements would test whether the morphology signal isolated here is the same cumulative structural transformation highlighted by recent S-PLUS studies.

Data availability

The input data are drawn from public SDSS spectroscopic and photometric tables, the MPA-JHU value-added catalogues, Galaxy Zoo classifications, the Simard et al. (2011) structural catalogue (VizieR `J/ApJS/196/11`), and the public group and compact-group catalogues cited in Sect. 2. The derived galaxy and group tables from this analysis will be made available through the CDS at publication. The full analysis and paper-generation code is available at https://github.com/MatthieuTricottet/CG_Gals; the release corresponding to this article will be archived with a DOI (e.g. Zenodo) at acceptance.

Acknowledgements. This publication uses data generated via the Galaxy Zoo project; we thank the many volunteers whose classifications made this analysis possible. This work made use of the MPA-JHU value-added spectroscopic catalogues, and of the Simard et al. (2011) structural catalogue obtained through the VizieR catalogue access tool (CDS, Strasbourg; catalogue `J/ApJS/196/11`). This research made use of Astropy (Astropy Collaboration et al. 2022), Astroquery (Ginsburg et al. 2019), NumPy (Harris et al. 2020), SciPy (Virtanen et al. 2020), pandas (McKinney 2010), Matplotlib (Hunter 2007), statsmodels (Seabold & Perktold 2010), and scikit-learn (Pedregosa et al. 2011).

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Appendix A: SDSS data query

The SQL query used to retrieve the spectral and photometric quantities from SDSS is listed below.

```sql
SELECT
    s.specObjID,
    s.z,
    p.petroMag_r - p.extinction_r as r_obs,
    p.petroMag_u - p.extinction_u as u_obs,
    p.petroMag_g - p.extinction_g as g_obs,
    p.petroMag_i - p.extinction_i as i_obs,
    p.petroMag_z - p.extinction_z as z_obs,
    p.objID,
    g.sfr_tot_p50, g.specsfr_tot_p50, g.lgm_tot_p50,
    l.h_alpha_eqw, l.h_beta_eqw, l.oiii_5007_eqw, l.nii_6584_eqw,
    l.h_alpha_flux, l.h_beta_flux, l.oiii_5007_flux, l.nii_6584_flux,
    z.p_el_debiased AS p_E,
    z.p_cs_debiased AS p_S
FROM SpecObj AS s
    JOIN PhotoObj AS p ON s.bestObjID = p.objID
    JOIN galSpecExtra as g ON s.specObjID = g.specObjID
    JOIN galSpecLine as l ON s.specObjID = l.specObjID
    JOIN zooSpec AS z ON s.specObjID = z.specObjID
WHERE s.z BETWEEN 0.005 AND 0.0452
    AND (p.petroMag_r - p.extinction_r <= 17.77)
    AND s.class = 'GALAXY'
    AND g.lgm_tot_p50 > -1000
```

Listing 1. Query used for selecting spectral & photometric data from the SDSS

Appendix B: sSFR classification details

We identify BPT-classifiable galaxies by requiring measured values of both emission-line ratios

$$\log_{10}([\mathrm{N\,II}]\lambda6584/\mathrm{H}\alpha) \quad \text{and} \quad \log_{10}([\mathrm{O\,III}]\lambda5007/\mathrm{H}\beta).$$

Among these galaxies, we flag AGN-like systems if they satisfy either

$$\log_{10}\frac{[\mathrm{N\,II}]}{\mathrm{H}\alpha} > 0,$$

or if they lie above the empirical demarcation of Kauffmann et al. (2003a):

$$\log_{10}\frac{[\mathrm{O\,III}]}{\mathrm{H}\beta} > \frac{0.61}{\log_{10}([\mathrm{N\,II}]/\mathrm{H}\alpha) - 0.05} + 1.3,$$

in which case they are excluded from the GMM calibration sample. This calibration flag is deliberately more inclusive than the three-class BPT scheme of Sect. 5.7, in which composite galaxies lie between the Kauffmann et al. (2003a) and Kewley et al. (2001) demarcations and only galaxies above the Kewley et al. (2001) line are labelled AGN-like, because its sole purpose is to keep possible AGN contamination out of the sSFR calibration. Galaxies with missing diagnostic line ratios are BPT-unclassified, not proven non-AGN. They are retained for the broad sSFR status assignment when valid SDSS stellar mass and sSFR estimates are available, while the emission-line/AGN-like analysis treats BPT-unclassified objects separately. The BPT-classifiable subset is shown in Fig. B.1.

Galaxies with `specsfr_tot_p50` equal to −9999 in the SDSS have no valid sSFR estimate; they are treated as missing data throughout: excluded from the GMM calibration, never assigned an sSFR class, omitted from every sSFR figure, and reported as counts only (Sect. 2.2). Galaxies with a measured sSFR are divided into quenched and star-forming classes with a two-component Gaussian Mixture Model (GMM) in the $\log M_\star$-$\log \mathrm{sSFR}$ plane (McLachlan & Peel 2000). The calibration excludes galaxies flagged as AGN-like where the required BPT line ratios are available, while BPT-unclassified galaxies with valid SDSS stellar mass and sSFR estimates can still receive a broad sSFR class. The quenched/star-forming boundary is the locus where the two component posterior probabilities are equal. A fixed-threshold robustness check confirms that the qualitative hierarchy is not tied to the exact GMM split.

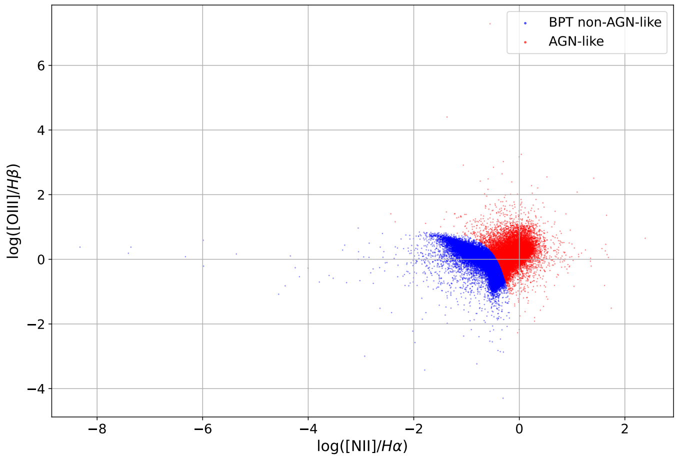

Fig. B.1. BPT diagram for galaxies with measured diagnostic line ratios. The AGN-like region is excluded from the sSFR calibration sample; galaxies without the required ratios are BPT-unclassified rather than classified as non-AGN by this diagram.

Appendix C: Supplementary diagnostic figures

This appendix collects the secondary diagnostic figures referenced in the main text.

C.1. sSFR and main-sequence diagnostics

Figures C.1–C.4 show the residual-sSFR distribution, the property distributions split by sSFR class (for all galaxies and for satellites by BGG class), and the polynomial-order comparison for the star-forming main sequence.

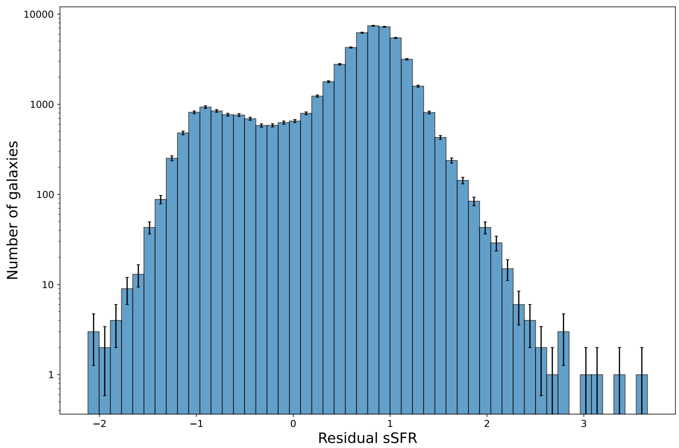

Fig. C.1. Distribution of residual sSFR offsets from the adopted star-forming main sequence.

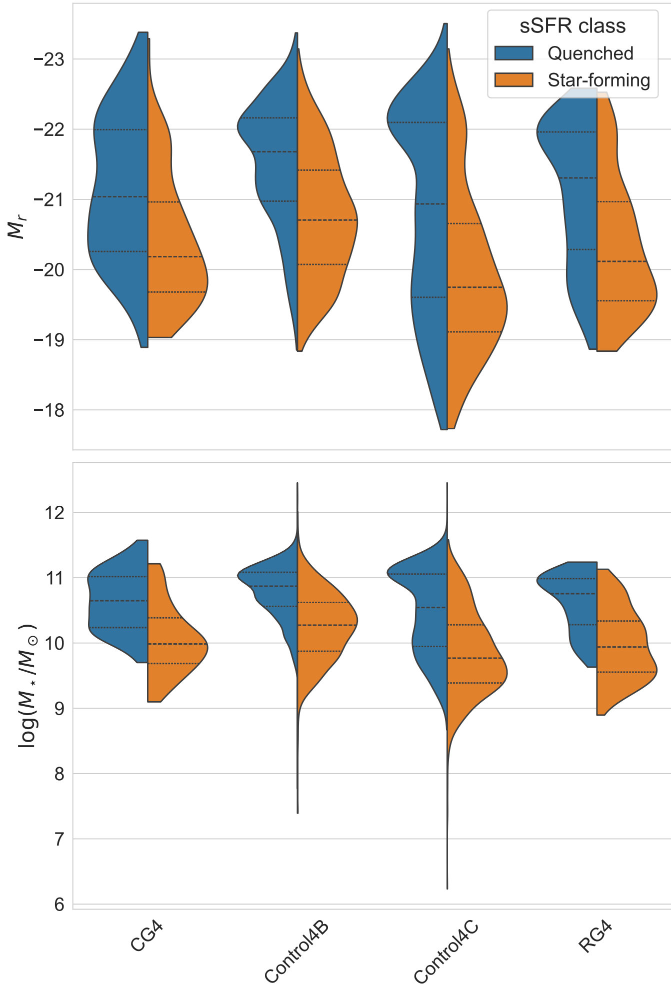

Fig. C.2. Distributions of various quantities for galaxies according to their sSFR class.

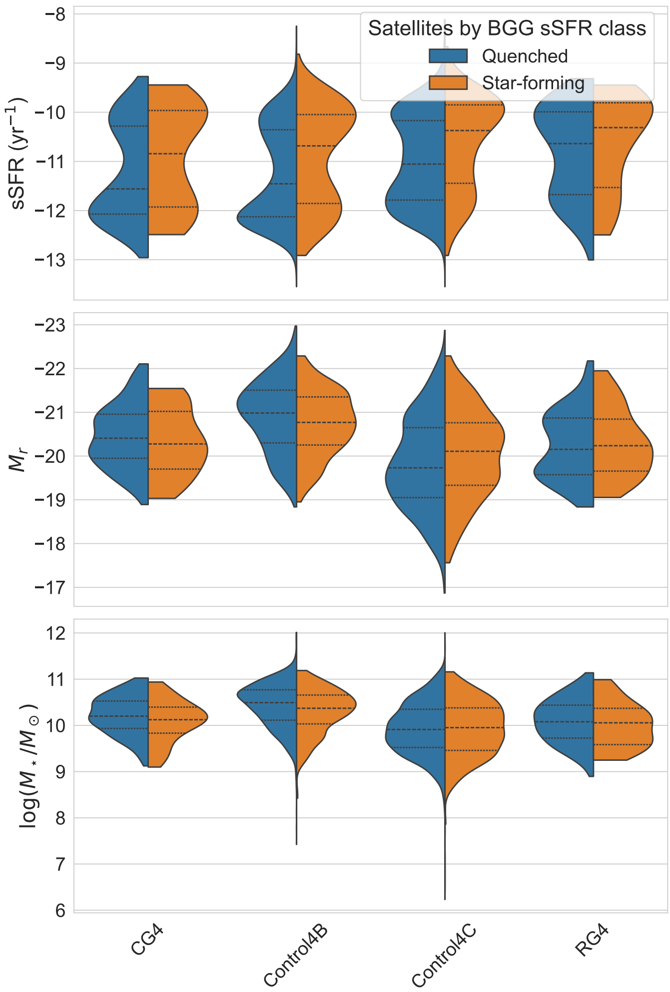

Fig. C.3. Distributions of various quantities for satellites according to the BGG sSFR class.

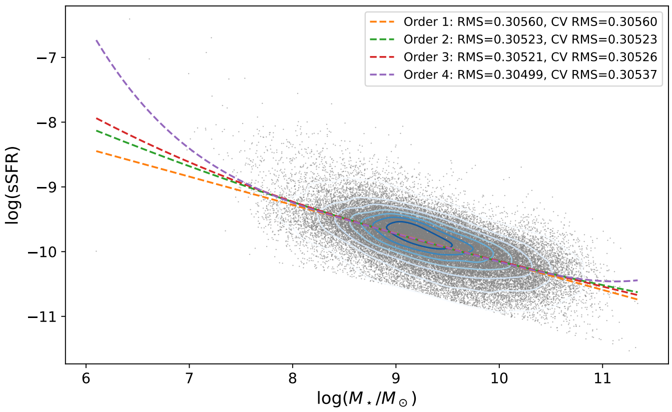

Fig. C.4. Polynomial fits of orders 1–4 to the star-forming main sequence. The legend reports residual RMS and five-fold cross-validated RMS scatter in dex; order 2 is adopted because it is the lowest-order model with the smallest cross-validated RMS scatter.

C.2. Group-scale and radial diagnostics

Figures C.5 and C.6 show the sSFR–distance relations, Figs. C.7 and C.8 the domination diagnostics, and Fig. C.9 the crossing-time trend.

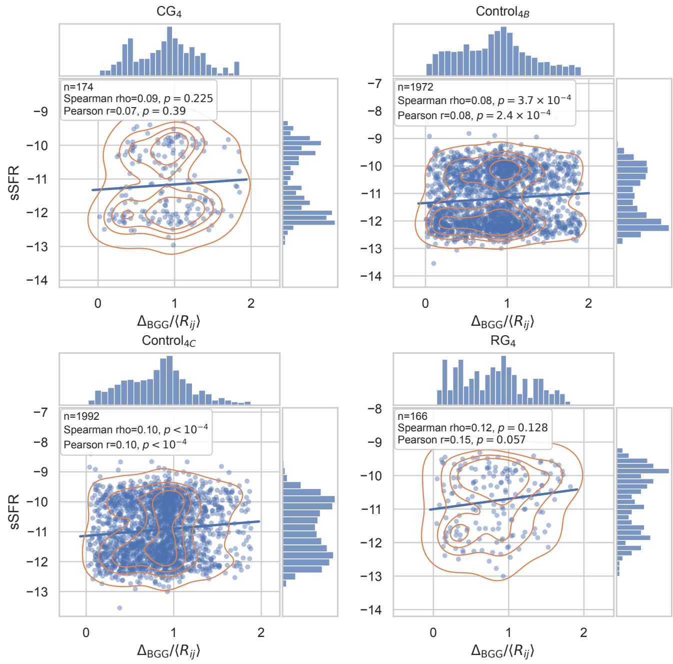

Fig. C.6. sSFR as a function of projected distance to the BGG normalized by $\langle R_{ij} \rangle$ for satellite galaxies in each sample.

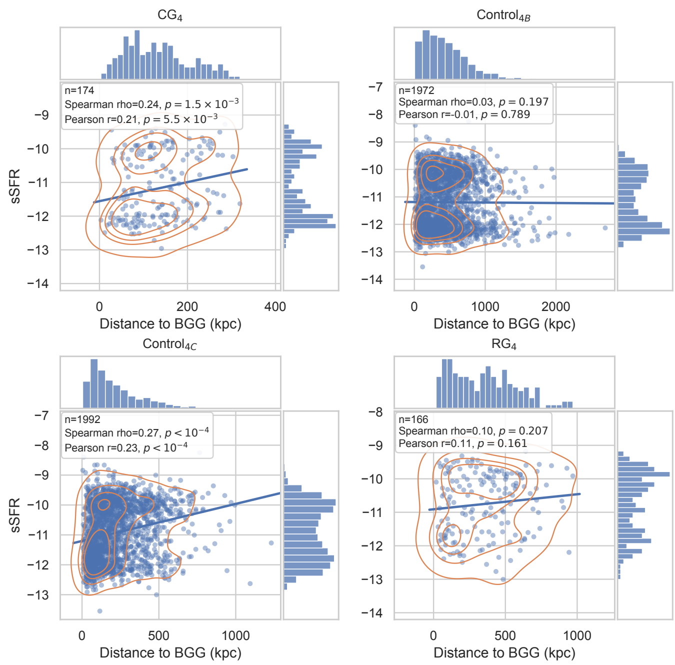

Fig. C.5. sSFR as a function of projected distance to the BGG, in kpc, for satellite galaxies in each sample.

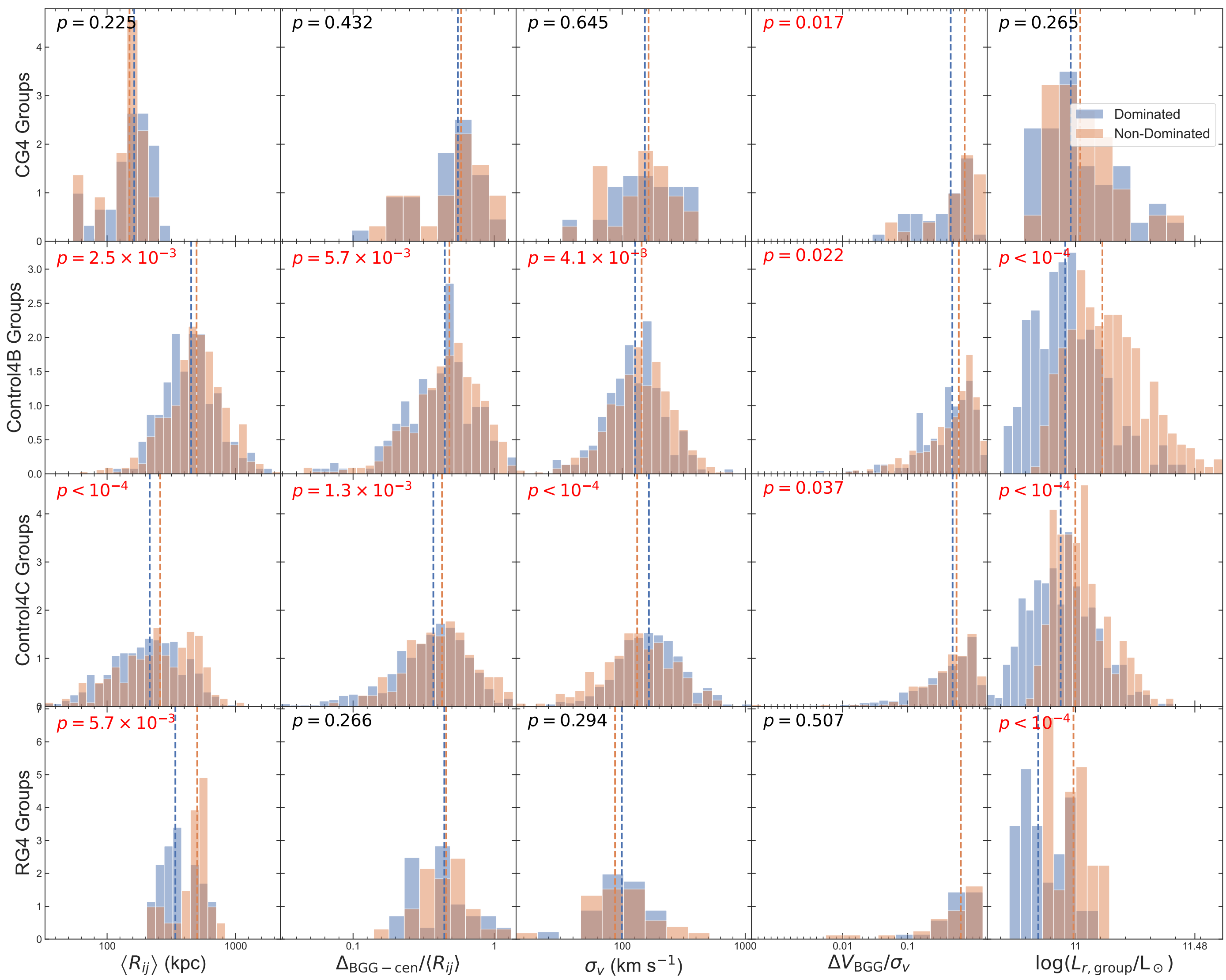

Fig. C.7. Distributions of the main group-scale observables in dominated and non-dominated systems.

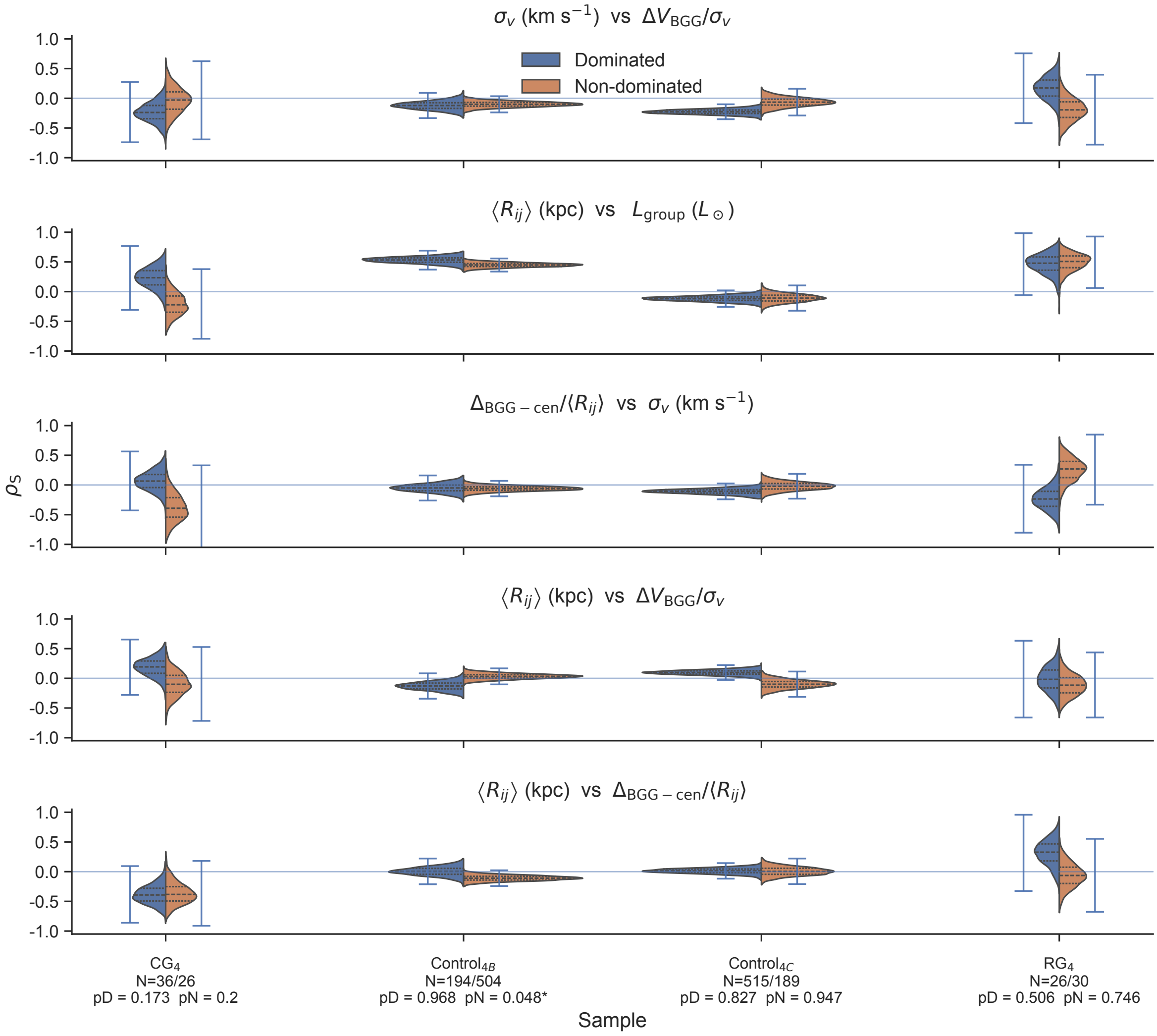

Fig. C.8. Bootstrap distributions of the Spearman coefficients for the group-property pairs selected by the XOR domination criterion, i.e. property pairs whose Spearman correlation, after a per-subsample Benjamini–Hochberg adjustment across pairs, is significant ($p_{\mathrm{adj}} < 0.10$) in exactly one of the dominated and non-dominated subsamples of at least one sample.

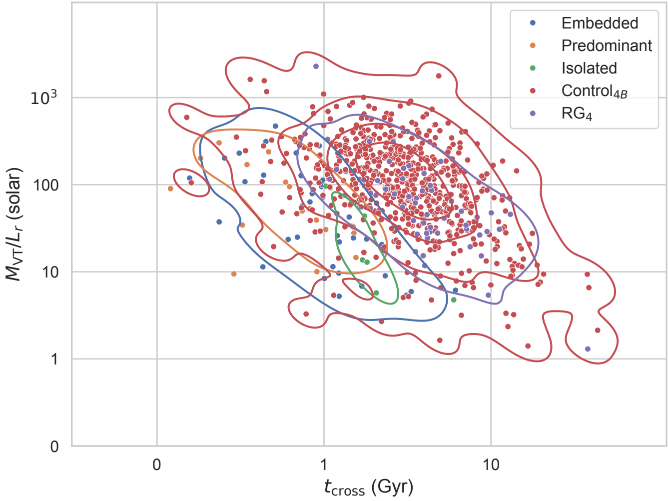

Fig. C.9. Crossing time versus virial mass-to-light ratio for the compact groups and their controls.

C.3. Colour diagnostics

Figures C.10–C.12 show the colour–mass relations of the catalogues, the satellite colour–mass comparison, and the mass- and redshift-adjusted colour residuals versus BGG-centric distance.

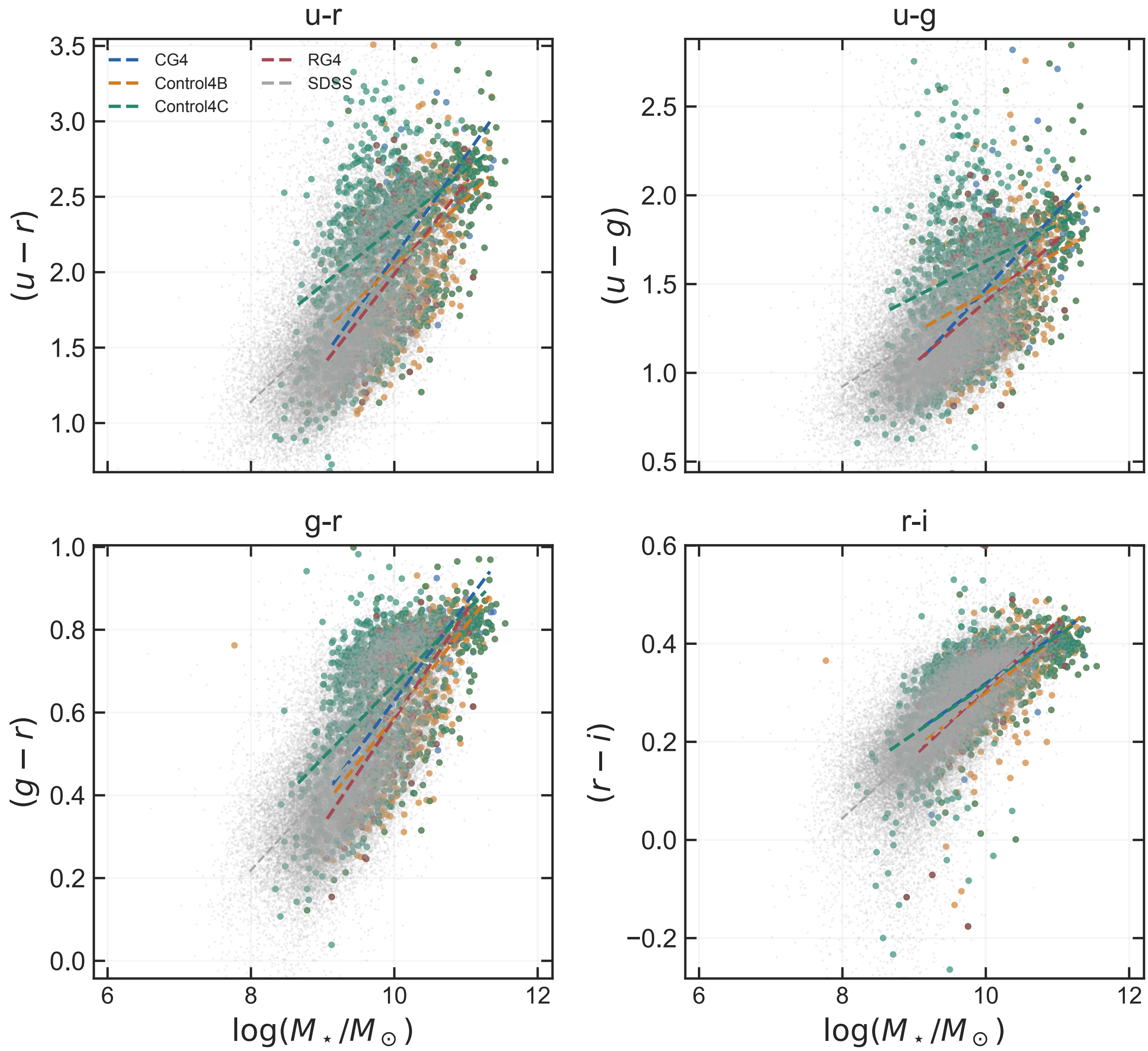

Fig. C.10. Optical colour as a function of stellar mass for CG_4, the three control catalogues, and the matched SDSS reference sample. Dashed lines show the fitted catalogue relations.

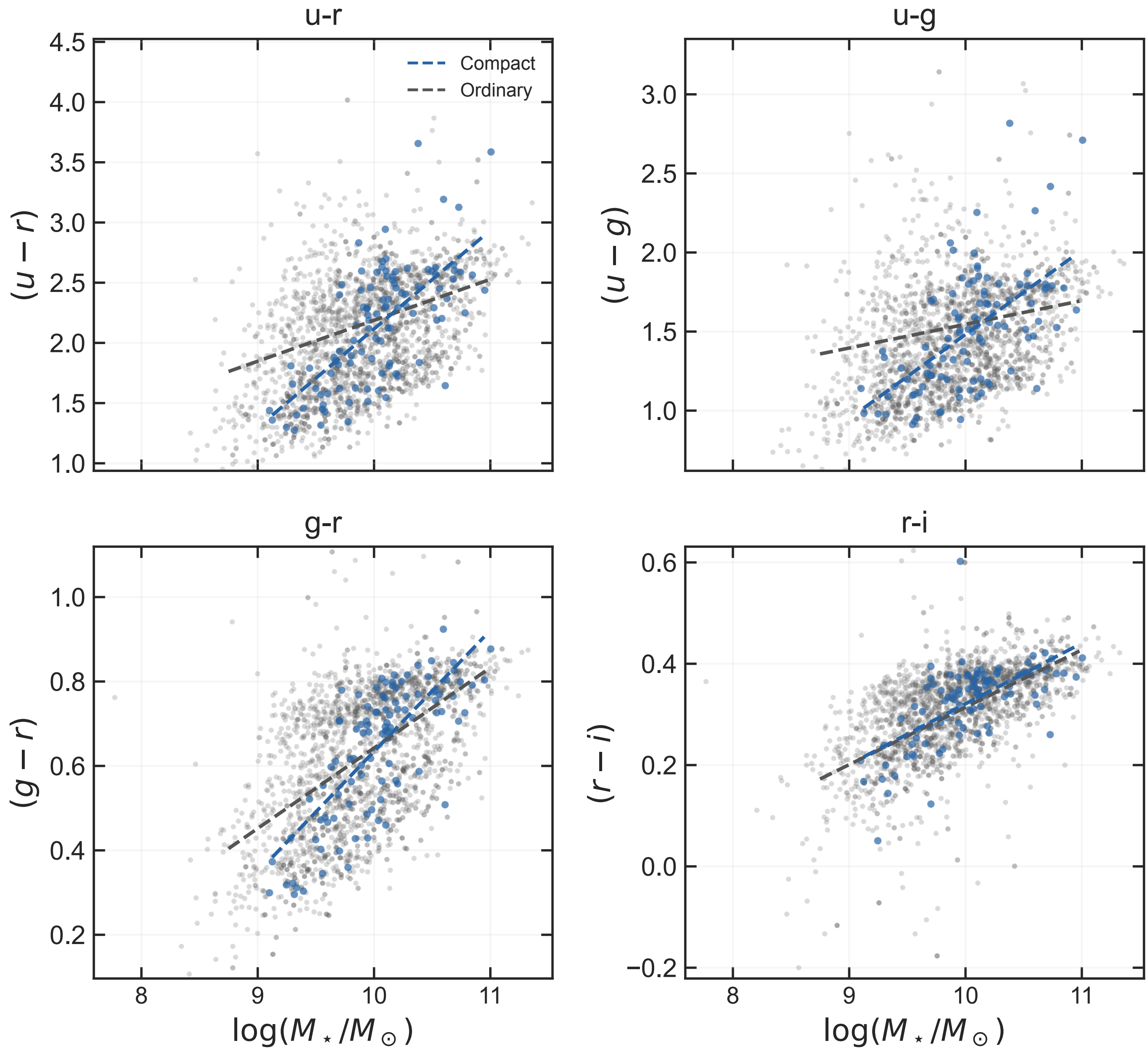

Fig. C.11. Satellite colour–mass relations in compact and ordinary groups. The fitted lines show both colour offsets at the reference mass and differences in slope.

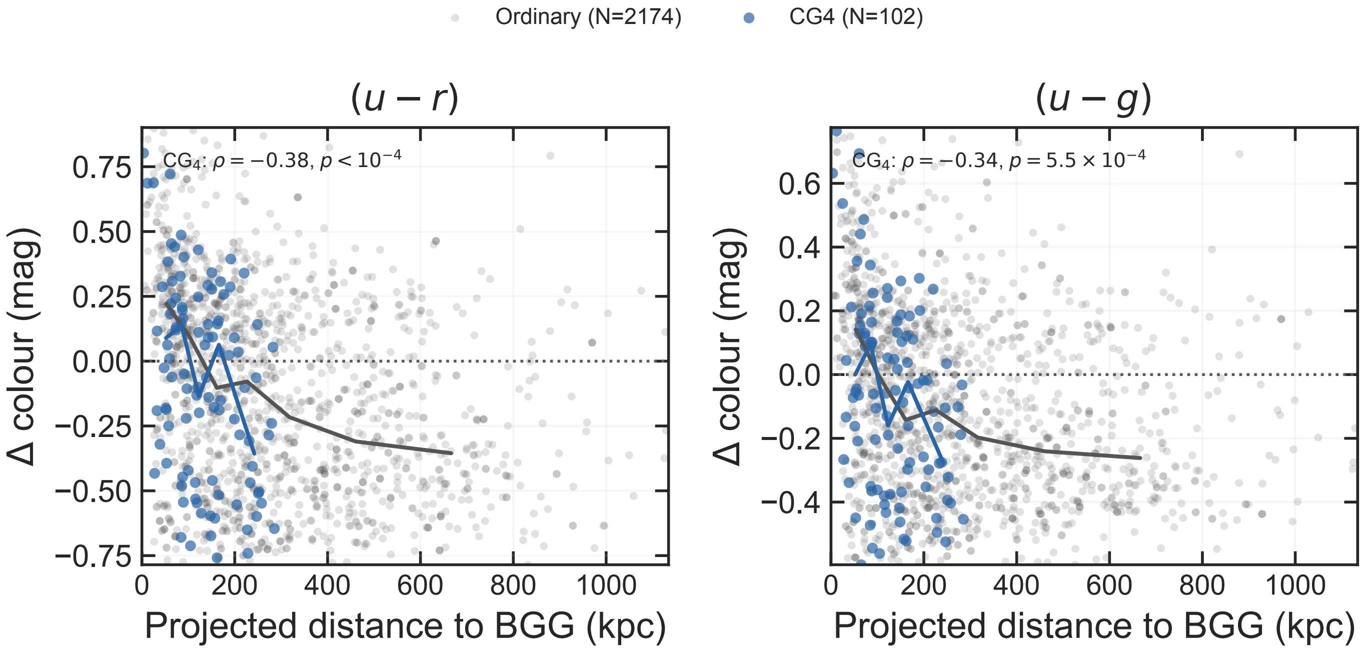

Fig. C.12. Mass- and redshift-adjusted satellite $(u-r)$ and $(u-g)$ residuals as a function of projected distance to the BGG. Lines connect binned medians for CG$_4$ and the pooled ordinary-group satellites; extreme plotting tails are omitted for readability, while the reported rank correlations use the full matched sample. More negative residuals are bluer than the ordinary-satellite reference relation.

C.4. Size diagnostics

Figures C.13–C.18 show the availability audit of the two size measures, the mass–size relations, the per-control size coefficients, the Simard–Petrosian measure difference, the R_e–Sérsic-n plane, and the radial size-residual diagnostic.

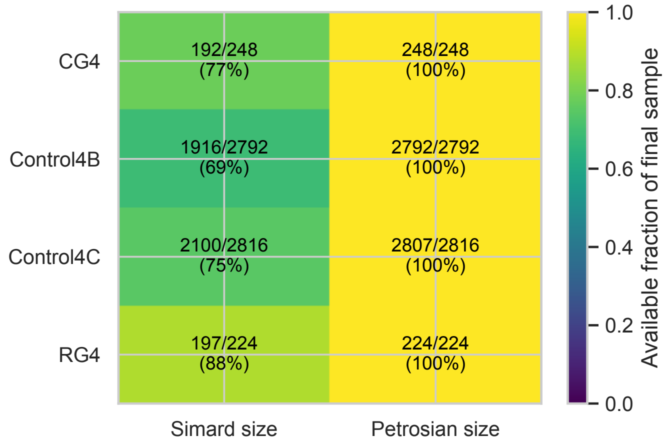

Fig. C.13. Availability of the seeing-corrected Simard half-light radius and the Petrosian half-light radius in each sample, after the bridge, quality, and blending cuts of Sect. 2.4.

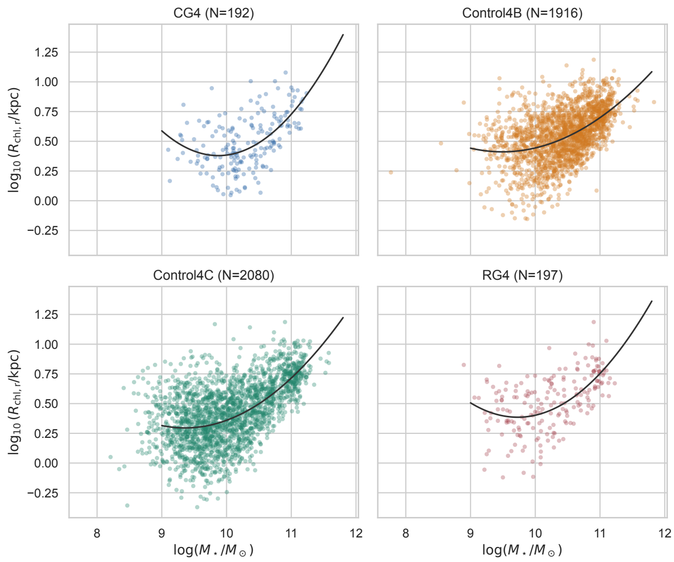

Fig. C.14. Seeing-corrected half-light radius versus stellar mass in each sample. Solid curves show the descriptive quadratic fits used to quote a predicted size at the reference stellar mass.

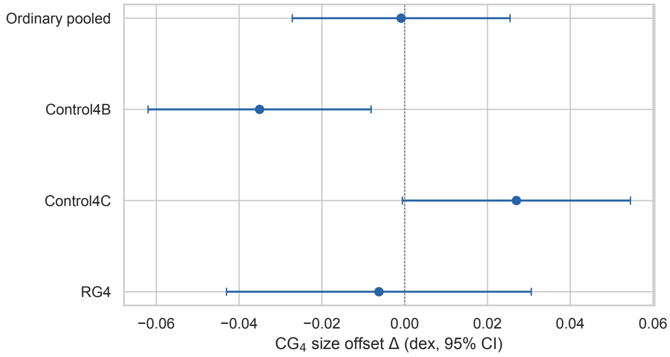

Fig. C.15. Mass- and redshift-adjusted CG_4 size coefficients against the pooled controls and against each control catalogue separately. Bars show 95% confidence intervals; negative values mean smaller CG_4 sizes at fixed mass.

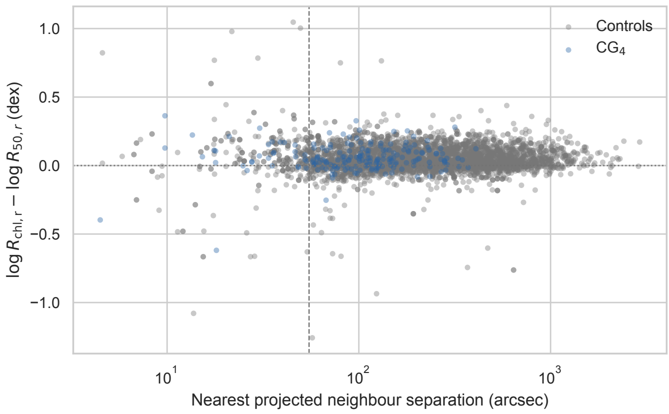

Fig. C.16. Per-galaxy difference between the Simard and Petrosian log half-light radii as a function of nearest projected neighbour separation. The dashed line marks the 55-arcsec crowding scale.

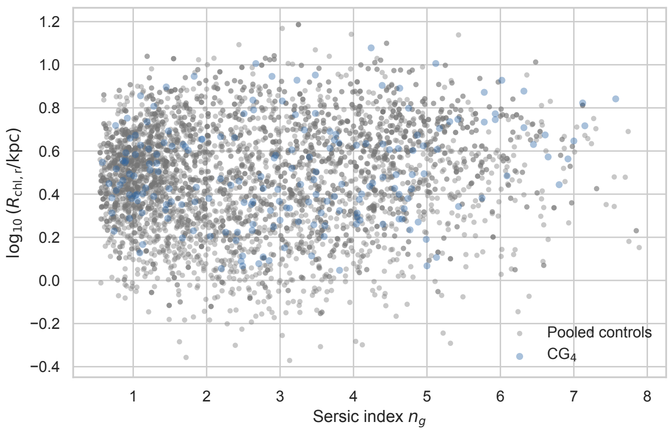

Fig. C.17. Half-light radius versus Sérsic index for CG_4 and the pooled controls, the plane used by the S-PLUS transition-galaxy studies.

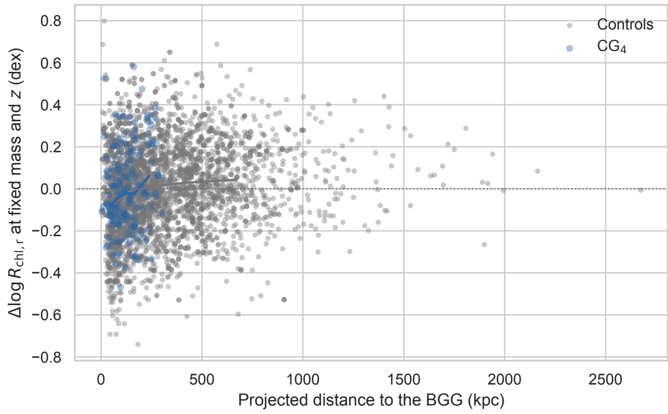

Fig. C.18. Mass- and redshift-adjusted satellite size residuals versus projected distance to the BGG. Lines connect binned medians for CG_4 and the pooled controls.

C.5. Robustness and selection diagnostics

Figures C.19–C.25 show the projected phase-space bins, the magnitude-gap diagnostics, the Hα equivalent-width distributions, the AGN-like fractions, the availability audit, and the colour-selection mass audit.

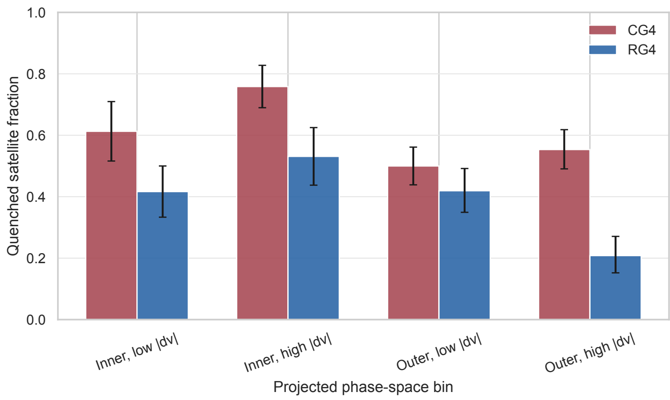

Fig. C.19. Quenched satellite fraction in coarse projected phase-space bins. Inner and outer refer to projected-distance rank from the BGG; low and high velocity are split at $|\Delta v_{\rm BGG}|/\sigma_v = 1$.

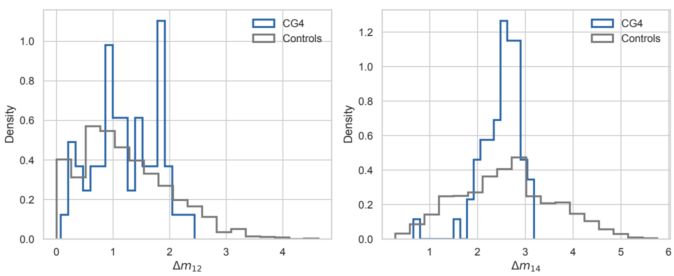

Fig. C.20. Distributions of the first–second and first–fourth magnitude gaps in CG$_4$ and the pooled controls.

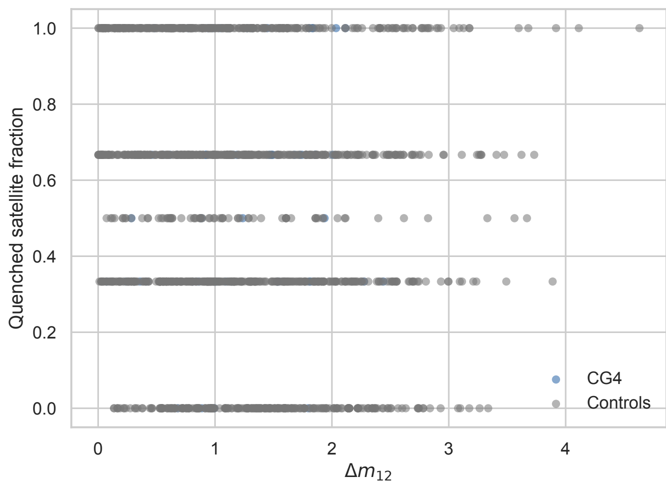

Fig. C.21. Quenched satellite fraction as a function of the first–second magnitude gap.

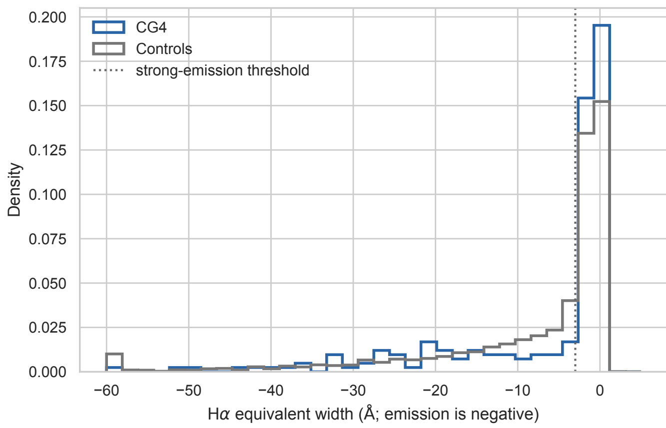

Fig. C.22. Hα equivalent-width distributions used for the limited recent-quenching diagnostic.

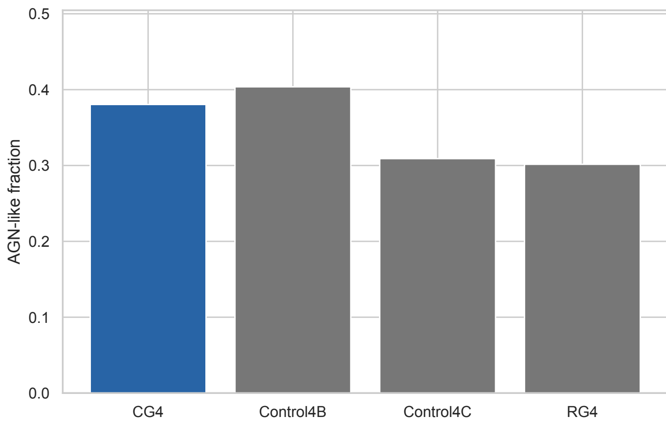

Fig. C.23. AGN-like fractions among galaxies with BPT-classifiable emission-line ratios. BPT-unclassified galaxies are not counted as non-AGN in this diagnostic.

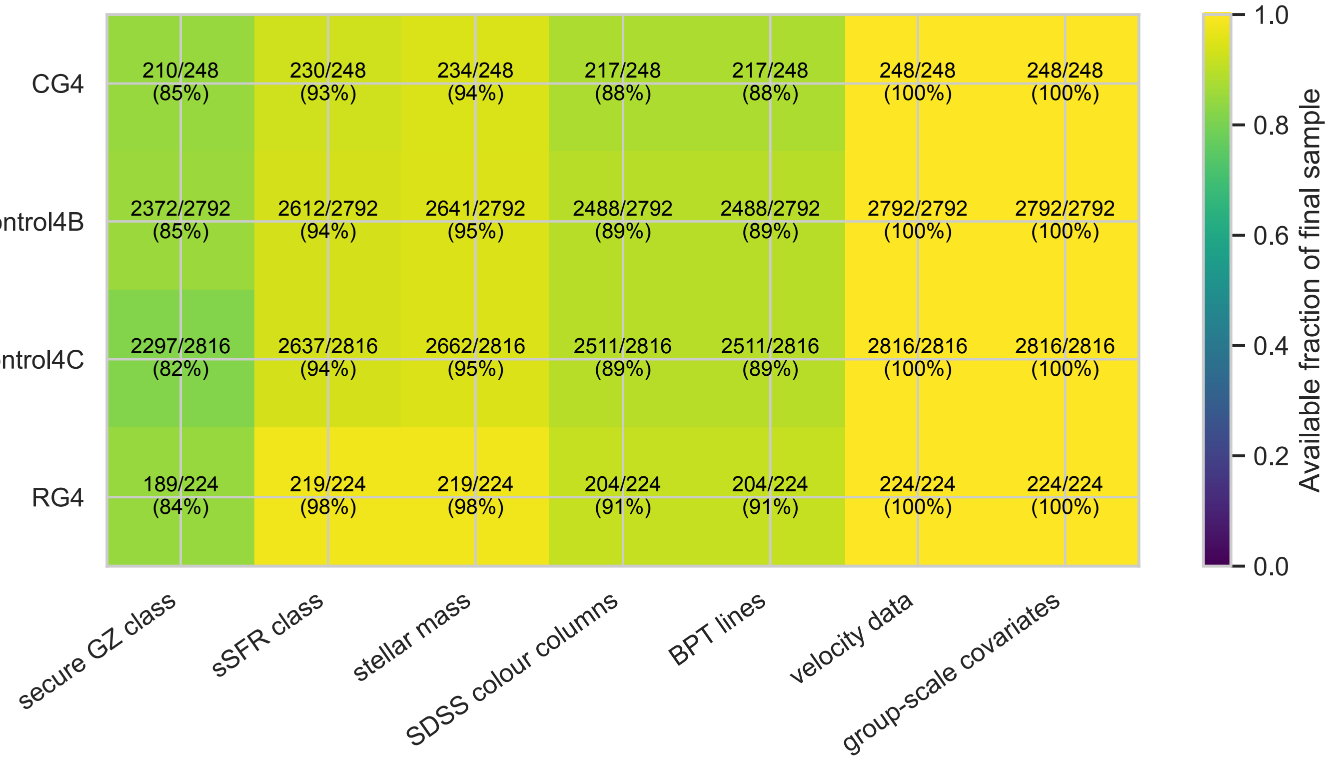

Fig. C.24. Availability fractions for the major derived quantities in each sample. The colour row records completeness of the four derived SDSS colour columns in the final catalogue, rather than the stricter colour-analysis matched subset used in Sect. 4.6.

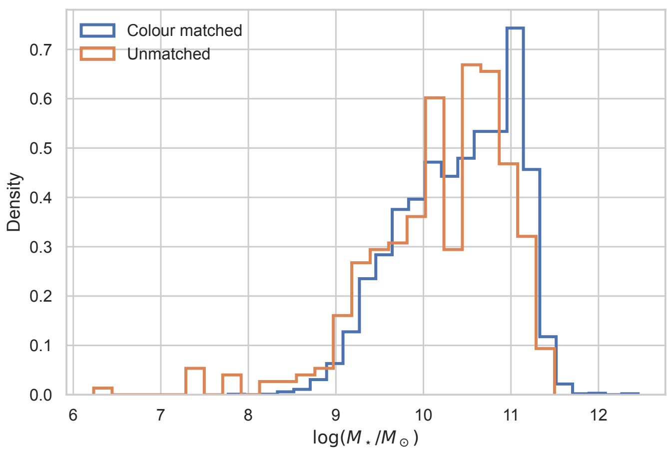

Fig. C.25. Stellar-mass distributions of galaxies with and without complete matched colour measurements.